

Production, Release and Transport of Noble Gases in the Continental Crust

Chris J. Ballentine

*Department of Earth Sciences
The University of Manchester
Manchester, M13 9PL, United Kingdom
cballentine@fs1.ge.man.ac.uk*

Pete G. Burnard

*Division of Geological and Planetary Sciences
MS 100-23, California Institute of Technology
Pasadena, California 91125*

INTRODUCTION

Noble gases within the crust originate from three main sources: the atmosphere, introduced into the crust dissolved in groundwater; the mantle, in regions of magmatic activity; and those produced **in the crust by the result of radioactive decay processes**. The continental crust contains approximately 40% of the terrestrial radioelements (Rudnick and Fountain 1995) that produce noble gases and, after the mantle and the atmosphere, forms the third major terrestrial noble gas reservoir (neglecting the core). In addition to these sources, contributions from interplanetary dust particles (IDP), cosmic ray interaction with the crustal surface and anthropogenic noble gases can in some cases be a significant source of noble gases in crustal materials. The use of noble gases to understand the role of fluids in different geological settings relies on their low natural abundance and chemical inertness. The low abundance of noble gases in crustal systems and their distinct isotopic character means that contributions from these different sources can often be resolved and quantified. With this, information is gained about the source of associated fluids, the environment from which they originated the physical manner in which they have been transported to the sampling site and the different phases that may have interacted within the crustal fluid system. This is only possible, however, with a detailed understanding of the processes that control the concentration and isotopic composition of the noble gases in different crustal environments.

The first part of this chapter deals with the three different mechanisms of noble gas production within the crust—radiogenic, nucleogenic, fissiogenic (Fig. 1). We show how production ratios are affected not only by the source region radioelement concentration, but in the case of nucleogenic reactions, also by the spatial distribution and concentration of the target elements. For completeness we consider cosmogenic noble gas production rates and Interplanetary Dust Particle (IDP) accumulation. We consider the total crustal budget and quantify how much of these differently sourced noble gases have, or can, contribute to the crustal fluid system. We then discuss how release from the minerals in which they are produced occurs by recoil, diffusion, fracturing and mineral alteration. The character of the mineral in which they are produced, the release process, the thermal regime and the ability of the surrounding fluid regime to transport the released noble gases, all play a role in fractionating the crustal noble gas elemental pattern and in determining their respective flux from the deeper crust into shallow systems. We detail how and where magmatic noble gases are introduced into the crust and discuss the relationship of ^4He and $^3\text{He}/^4\text{He}$ with heat flow.

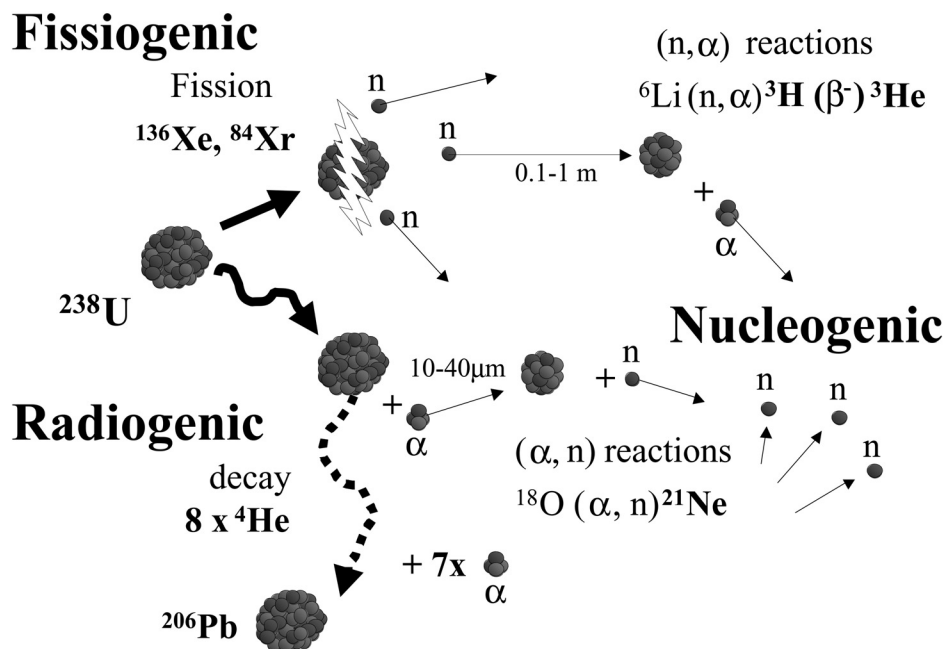


Figure 1. Schematic diagram of ^{238}U decay to ^{206}Pb , showing fission and α decay. Fission results in the direct formation of heavy Kr and Xe isotopes and neutrons. Most α particles stabilise to become ^4He , although a small portion react directly with local ($0-40\text{ }\mu\text{m}$) light nuclei in (α, n) reactions to create nucleogenic noble gases such as ^{21}Ne in the reaction $^{18}\text{O}(\alpha, n)^{21}\text{Ne}$. Less than 20% of the subsurface neutron flux is produced directly by fission (e.g., Yatsevich et al. 1997). The remainder is dominated by production from the (α, n) reactions. The neutrons in turn also react with nuclei on a $0.1-1\text{ m}$ length scale. Typical reactions include the (n, α) route producing, for example, ^3He in the reaction $^6\text{Li}(n, \alpha)^3\text{H}(\beta^-)^3\text{He}$. The nuclei available for reaction in the $0-40\text{ }\mu\text{m}$ range of an α -emitter determine the production rate of (α, n) -derived noble gases. The greater length-scale of the neutron penetration distance means that noble gases produced by the (n, α) route are not sensitive to mineral-scale elemental heterogeneity.

RADIOGENIC, NUCLEOGENIC, AND FISSIOGENIC NOBLE GASES

The subsurface neutron flux and reaction probability

Neutron interaction with atomic nuclei is directly linked to the production of several noble gas isotope species in the crust. Before considering the production of crustal noble gases it is essential to first understand the factors controlling the subsurface neutron flux. There are three main types of reaction that produce neutrons in the crust: Cosmic ray interactions; spontaneous fission; and alpha particle interaction with light nuclei. Cosmic ray interaction is only important within the top few meters of the crust (Niedermann 2002, this volume) and we neglect this source of neutrons here.

Neutrons are produced by (α, n) reactions within $0-40\text{ }\mu\text{m}$ of $^{235,238}\text{U}$ and ^{232}Th , together with a small contribution from the spontaneous fission of ^{238}U to give:

$$n = (\alpha, n)\text{U} + (\alpha, n)\text{Th} + (\text{sf}, n)\text{U} \quad (1)$$

where n is the total number of neutrons and $(\alpha, n)\text{U}$, $(\alpha, n)\text{Th}$, $(\text{sf}, n)\text{U}$ are the neutrons produced by $^{235,238}\text{U}$, ^{232}Th (α, n) reactions, and spontaneous fission of ^{238}U respectively. Spontaneous fission neutron contributions from ^{232}Th and ^{235}U are negligible (Andrews and Kay 1982; Morrison and Pine 1955). The spontaneous fission neutron yield of ^{238}U is 2.2 ± 0.3 neutrons per fission (Morrison and Pine 1955), and the decay constant for spontaneous ^{238}U fission is $8.6 \times 10^{-17}\text{ yr}^{-1}$ (Eikenberg et al. 1993). This compares with the ^{232}Th neutron yield of 2.5 neutrons per fission with a fission decay constant estimated

to be $<2 \times 10^{-21} \text{ yr}^{-1}$ (Wieler and Eikenberg 1999). The spontaneous fission neutron production rate is calculated by:

$$(\text{sf},n) = (N_A/A_U) \times \lambda_f \times \text{yield} \times [U] \times 10^{-6} \quad (2)$$

where N_A = Avogadro's number (6.023×10^{23}), A_U = molar mass of uranium, λ_f = fission decay constant and $[U]$ = the concentration of U in ppm to give

$$(\text{sf},n)U = 0.4788 [U] (\text{g}^{-1} \text{ yr}^{-1}) \quad (3)$$

for a natural uranium composition of $^{235}\text{U} = 0.72\%$ and $^{238}\text{U} = 99.28\%$. It should be noted that ^{238}U has a very high resonance capture energy at $\sim 10\text{eV}$, and in the case where these neutrons are produced within accessory phase minerals containing high concentrations of U, such as uraninite and monazite, the number of neutrons available for reaction is only between 0.2 and 0.4 times the amount produced (Gorshkov et al. 1966).

The neutron yield from (α,n) reactions is a function of the energy dependant (α,n) reaction cross section (Feige et al. 1968) and the rate of energy loss of the α -particles and reaction channels with the excited states of residual nuclei (Yatsevich and Honda 1997). The two important parameters determining the reaction cross section are the reaction energy, Q , and the height of the Coulomb barrier, B (Feige et al. 1968). B is defined by:

$$B = 2.8 Z / (1.4A^{1/3} + 1.2) \text{ MeV} \quad (4)$$

where Z and A are the atomic number and weight of the target nucleus respectively. The highest energy of a naturally emitted α -particle is 8.78 MeV and Coulomb interactions with the medium electrons will slow the α -particles before reaction takes place. Isotopes for which $B > 8 \text{ MeV}$ (elements with $Z > 16$) have a Coulomb barrier above the natural upper limit and are not therefore a significant source of subsurface neutrons. For an $m_1(\alpha,n)m_2$ reaction

$$Q = [(m_1 + m_\alpha) - (m_2 + m_n)] c^2 \quad (5)$$

where m_1 , m_α , m_2 and m_n are the masses of m_1 , the α -particle, m_2 and neutron respectively and c the speed of light. The reaction threshold for an endothermic reaction, E_{th} , is given by:

$$E_{\text{th}} = -[(m_1 + m_\alpha) / m_1] Q \quad (6)$$

and is the minimum kinetic energy an α -particle must have in order for the reaction to be energetically favorable. Even though the Coulomb barrier is reached, ^{16}O , ^{28}Si , which constitute some 75% of the Earth's crust by weight, as well as ^{40}Ca (also excluded by the Coulomb barrier) and ^{24}Mg do not therefore significantly participate in natural (α,n) reactions.

As demonstrated above, the reaction cross section is dependant on the energy of the reacting α -particle. To calculate the neutron production of a compound, it is necessary to know the neutron yield and the mass stopping ability of each element, the latter of which is also dependant on the α energy. Experimentally derived neutron yields exist for the range of naturally occurring α -particle energies (e.g., Feige et al. 1968; Jacobs and Liskien 1983; West and Sherwood 1982) and can be fitted to a polynomial to allow calculation of the yield at any alpha energy. From the neutron yield of the constituent elements, Y_i , the neutron yield of a particular compound, Y_c , can be calculated from

$$Y_c = \sum W_i S_i^m Y_i \quad (7)$$

where W_i and S_i^m are the mass fraction and mass stopping energy for alpha particles of element i . With the data supplied by Feige (1968), and the assumption that neutron

contributions from O and Ca can be neglected an empirical U and Th derived neutron flux for an homogenous media can be calculated following:

$$(\alpha,n)U = 0.01 [U] \{13.8[Na] + 5.4[Mg] + 5.0[Al] + 1.31[Si] + 2.0[C]\} g^{-1} yr^{-1} \quad (8)$$

$$(\alpha,n)Th = 0.01 [Th] \{6.0[Na] + 2.45[Mg] + 2.55[Al] + 0.56[Si] + 0.83[C]\} g^{-1} yr^{-1} \quad (9)$$

where [U] and [Th] are the respective concentration of U and Th in ppm whereas the concentration terms for Na, Mg, Al, Si and C are their percentage concentrations in the rock. The total subsurface neutron density, N, can be derived by combining Equations (1), (3), (8) and (9) to give

$$N(\text{neutrons } g^{-1} yr^{-1}) = 0.01[U] \{13.8[Na] + 5.4[Mg] + 5.0[Al] + 1.31[Si] + 2.0[C]\} + 0.01[Th] \{6.0[Na] + 2.45[Mg] + 2.55[Al] + 0.56[Si] + 0.83[C]\} + 0.4788 [U] \quad (10)$$

This is similar to the equation presented by Andrews (1985) but differs in neglecting Ca (see earlier discussion) instead of C as a significant neutron source.

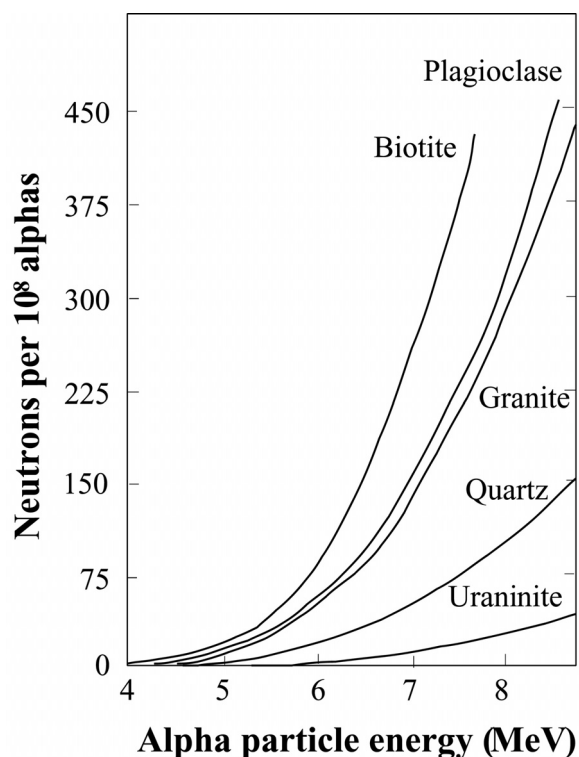


Figure 2. Martel et al. (1990) show the energy dependence of neutron production from (α,n) reactions for selected mineral and rock compositions using the data of Gorshkov et al. (1962). Neutron fluxes calculated assuming element homogeneity, would significantly overestimate the flux if the radioelements are concentrated in accessory phases such as uraninite. [Used by permission of Elsevier, from Martel et al (1990), *Chemical Geology*, Vol. 88, Fig. 5, p. 215.]

Using the light element neutron yield from Feige (1968) and the stopping energy distribution from Ziegler (1977), Martel et al. (1990) have calculated the α -energy-dependent neutron yield for selected mineral and rock compositions (Fig. 2). The results show the dependence of U and Th siting on the neutron flux produced in a rock, also showing that neutron production probability falls to zero at energies of <4.0 MeV. The effective neutron production range is therefore only 0-30 μm in major minerals and reduced to 0-20 μm in accessory phase minerals such as monazite and uraninite. Martel et

al. (1990) conclude that estimates of neutron yields based on a homogenous element distribution may significantly overestimate subsurface neutron flux if the bulk of the radioelements are within the accessory phases. The size distribution of the accessory phase is also critical. Figure 3 shows the calculated neutron production rate for spherical accessory phases in a biotite matrix as a function of their radii. As the accessory mineral radius approaches zero the production rate approaches that of biotite. As the mineral radius approaches the limit of the α -particle penetration distance of 20 μm , the neutron flux becomes dominated by the spontaneous fission contribution (Martel et al. 1990).

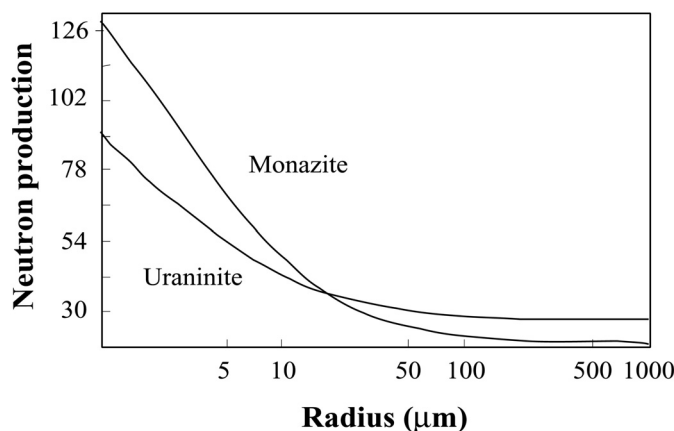


Figure 3. Martel et al. (1990, show the calculated neutron production rate as a function of grain size for spherical uraninite and monazite grains in a biotite matrix. The values are relative to an arbitrary value of 100 assigned to the yield from Al bombarded by mono-energetic 5.3 MeV alpha particles. The crossover in the curves at low neutron production rates is the result of the high U/Th ratio in uraninite relative to monazite and thus a larger contribution from spontaneous fission. [Used by permission of Elsevier, from Martel et al (1990), *Chemical Geology*, Vol. 88, Fig. 6, p. 217.]

Comparison of measured *in situ* and calculated neutron fluxes were originally made by Gorshkov et al. (1966) who showed a 20% agreement between measured and predicted values. More recently, Andrews (1986) and Chazal et al. (1998) have compared measured vs. predicted values for the Stripa granite, Sweden, and in the Modane underground laboratory, Switzerland respectively. The calculated flux in the Stripa granite is 15% higher than the measured value, again showing that for this case the assumption of element homogeneity is reasonable. In contrast, Chazal et al. (1998) use a Monte Carlo approach to calculate the neutron production rate, but measure a flux four times greater than calculated values. They ascribe this difference to the difficulty in determining the water content of the rock. The moderating capacity of water (its ability to efficiently absorb neutrons) was neglected in the calculations of Andrews (1986) and may have been overestimated by Chazal et al. (1998).

Neutrons produced by (α ,n) reactions have energies typically on the order of several MeV. Through elastic and inelastic scattering the neutron energy is reduced and, after many collisions, thermalizes to energies similar to gas molecules at the ambient temperature (e.g., ~ 0.03 eV at 25°C). Reaction probabilities for fast neutrons are generally much smaller than those for epithermal or thermal neutron reactions, particularly for light elements. Nevertheless, fast neutron reactions mean that only a portion (p_{th}) of the neutrons reach thermal energies. Morrison and Pine (1955) used an estimate for effective resonance cross section in a chain reacting nuclear pile to obtain $0.6 < p_{\text{th}} < 0.9$, with a best estimate of $p_{\text{th}} = 0.8$. Other workers argue that the high abundance of light elements (O, Si) in a typical rock matrix means that this may be an underestimate for the rock medium and, given other uncertainties, can be considered to be unity (Andrews 1985). Noble gas production rates involving reactions with thermal neutrons, such as the ${}^6\text{Li}(n,\alpha){}^3\text{H}(\beta^-){}^3\text{He}$ reaction, are

relatively simple to calculate for a given neutron flux. The fraction of neutrons in a rock matrix captured by element x is given by the equation:

$$F_x = \sigma_x N_x / \sum_i \sigma_i N_i \quad (11)$$

where σ and N are the neutron capture cross sections and abundances in moles, respectively, of all the elements present in the rock. The value of this summation is determined to within 1% by 17 dominant elements for all rock types (Andrews and Kay 1982) (Table 1).

Table 1. Thermal neutron capture probabilities for elements in rock matrices.

<i>NEUTRON CAPTURE PROBABILITY (MOL BARNS)</i>									
Cross-section (barns)		Ultra- mafic	Basalt	Granite	Sand- stone	Lime- stone	Lower* Crust	Middle* Crust	Upper* Crust
Li	71	5.00E-06	1.50E-04	4.00E-04	1.50E-04	3.40E-04	6.14E-05	7.16E-05	2.05E-04
B	755	7.00E-05	3.50E-04	1.04E-03	2.44E-03	1.40E-03	3.50E-04	3.50E-04	1.04E-03
Na	0.534	1.30E-04	4.50E-04	6.40E-04	7.00E-05	0.00E+00	4.48E-04	5.51E-04	6.72E-04
Mg	0.064	6.80E-04	1.20E-04	0.00E+00	0.00E+00	1.20E-04	1.13E-04	5.40E-05	3.49E-05
Al	0.232	4.00E-05	7.50E-04	6.60E-04	2.10E-04	4.00E-05	7.56E-04	7.06E-04	6.93E-04
Si	0.16	1.08E-03	1.37E-03	1.84E-03	2.09E-03	1.40E-04	1.39E-03	1.61E-03	1.76E-03
Cl	32.6	5.00E-05	5.00E-05	2.20E-04	1.40E-04	1.40E-04	5.00E-05	5.00E-05	2.20E-04
K	2.07	2.00E-05	4.40E-04	1.76E-03	5.60E-04	1.40E-04	2.64E-04	8.83E-04	1.49E-03
Ca	0.44	8.00E-05	7.40E-04	1.70E-04	4.20E-04	3.32E-03	7.38E-04	4.00E-04	3.30E-04
Ti	6.1	4.00E-05	1.15E-03	2.90E-04	1.90E-04	5.00E-05	6.11E-04	5.35E-04	3.82E-04
Cr	3.1	1.20E-04	1.00E-05	0.00E+00	0.00E+00	0.00E+00	1.28E-05	4.95E-06	2.09E-06
Mn	13.3	3.60E-04	4.80E-04	1.40E-04	1.00E-05	2.70E-04	1.87E-04	1.87E-04	1.50E-04
Fe	2.56	4.50E-04	3.90E-04	1.23E-03	4.40E-04	1.70E-04	2.69E-03	2.05E-03	1.44E-03
Co	37.5	1.30E-04	3.00E-05	0.00E+00	0.00E+00	0.00E+00	2.42E-05	1.59E-05	6.36E-06
Ni	4.54	1.50E-04	1.00E-05	0.00E+00	0.00E+00	0.00E+00	6.81E-06	2.55E-06	1.55E-06
Sm	5820	2.00E-05	1.90E-04	3.40E-04	3.80E-04	5.00E-05	1.08E-04	1.70E-04	1.74E-04
Gd	49000	1.60E-04	1.56E-03	2.80E-03	3.11E-03	4.00E-04	9.66E-04	1.25E-03	1.18E-03
<i>Total capture probability</i>		0.00359	0.00824	0.0115	0.0102	0.00658	0.00878	0.00889	0.00979

Table 1 is after Andrews and Kay (1982).

Elements that do not contribute >1% of the total capture probability for at least one rock type have not been tabulated.

Rock composition used is from Parker (1967).

*Average crust values from Rudnick and Fountain (1995); <http://earthref.org/GERM/>

The calculation of noble gas production rates involving fast neutron interactions is more complex, and requires a detailed calculation of the energy spectrum of the produced neutrons, and the rate and probability with which they thermalize. For example, the reactions $^{24}\text{Mg}(n,\alpha)^{21}\text{Ne}$ and $^{25}\text{Mg}(n,\alpha)^{22}\text{Ne}$ both have reaction energy thresholds in the range 2-4 MeV. Rison (1980) in calculating the Ne isotope yields from these reactions, estimated the neutron spectrum from the element (α,n) reactions as a δ function at the energy ($E_\alpha + Q$), where E_α is the energy of the impinging alpha particle and Q is the reaction energy release. The number of neutrons was estimated from experimentally determined yields of (α,n) reactions. Yatsevich and Honda (1997) note that this approach does not take into account α -particle energy loss before interaction and ignored reaction channels with the excited states of residual nuclei. Consequently the calculated Ne isotope yields are significantly exaggerated. Yatsevich and Honda (1997) rectify this by estimating the energy spectrum of neutrons from the thick-target angle-integrated spectra of neutrons emitted in (α,n) reactions on light elements measured by Jacobs and Liskien

(1983). This data is for the energy interval 4.0-5.5 MeV, and has been extended by fitting the data to a constant form function and incremental extrapolation to 9.0 MeV to enable interpolation over the different natural α -energy intervals. Combining this approach with a neutron transport model incorporating both elastic and inelastic scattering plus residual nuclei scattering results in a neutron spectrum with a significantly lower average energy than the earlier work of Rison (1980) (Fig. 4). This in turn results in ^{21}Ne and ^{22}Ne yields by the ^{24}Mg and ^{25}Mg reactions calculated by Yatsevich and Honda (1997) being approximately five times lower than the earlier values of Rison (1980).

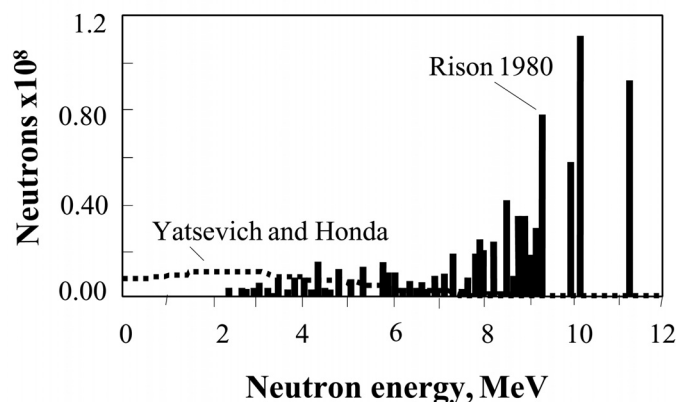


Figure 4. Yatsevich and Honda (1997) calculate the energy spectrum of neutrons produced from U and Th (α, n) reactions in an homogenous mantle over 4.5 Ga. They calculate a lower energy spread than the simplified calculation of Rison (1980) because of the inclusion of the effect of elastic and, to a lesser extent, inelastic scattering of the neutrons. Calculated yields for fast neutron reactions, such as with $^{24,25}\text{Mg}$ that have a 2-4 MeV threshold, are therefore much lower with the revised neutron energy spectrum. [Used by permission of the American Geophysical Union, from Yatsevich and Honda (1997), *Journal of Geophysical Research*, Vol. 104, Fig. 2, p. 10294.]

Helium

The present day ^4He production in the crust is dominated by the α -decay of the $^{235,238}\text{U}$ and ^{232}Th decay chains, and is therefore directly proportional to the concentration of these radioelements in the crust. Although some α -particles are produced by a variety of crustal nuclear reactions, these sources of ^4He production are many orders of magnitude smaller and can be neglected; similarly α -particles consumed by (α, n) reactions within the crust are a very small proportion of the α -particles produced. The ^4He produced from each radio-isotope, R, expressed as a function of present day concentration in the rock is given by

$$^4\text{He atoms g}^{-1} \text{ yr}^{-1} = X_r [R] (N_A/A_r) \times 10^{-6} (e^{\lambda_r t} - 1) \times \text{yield}_r \quad (12)$$

where X_r = fractional natural abundance of isotope R, N_A = Avogadro's number (6.023×10^{23}), A_r = molar mass of R (g), λ_r = decay constant of R (yr^{-1}), yield = number of α particles emitted in the complete decay chain, $[R]$ = the concentration of R in ppm and t = age (yr). For ^{238}U , ^{235}U and ^{232}Th , $\lambda_{238} = 1.55 \times 10^{-10}$, $\lambda_{235} = 9.85 \times 10^{-10}$, $\lambda_{232} = 4.95 \times 10^{-11}$ (Steiger and Jäger 1977), with an α yield in each decay chain of 8, 7 and 6 and a natural abundance of $X_{238} = 0.9928$, $X_{235} = 0.0072$ and $X_{232} = 1.000$ respectively, the number of atoms of ^4He produced in 1 gram of rock per year becomes

$$^4\text{He atoms g}^{-1} \text{ yr}^{-1} = (3.115 \times 10^6 + 1.272 \times 10^5) [\text{U}] + 7.710 \times 10^5 [\text{Th}] \quad (13)$$

^3He production within the crust is dominated by thermal neutron capture by ^6Li in the reaction $^6\text{Li}(n, \alpha)^3\text{H} (\beta^-)^3\text{He}$. Other reactions are detailed in Mamyrin and Tolstikhin (1984, p. 101), but for average crustal compositions, these have a ^3He yield at least four

orders of magnitude lower than the ${}^6\text{Li}$ route and are neglected here. The rate of ${}^3\text{He}$ production therefore, is directly proportional to the subsurface thermal neutron density (Eqn. 10), the number of neutrons that reach thermal energy level ($P_{\text{th}} = 0.8$) and the relative capture cross section of Li (Eqn. 11). Taking the composition of a homogenous average upper crust (Table 2) as a working example, we obtain:

$$\begin{aligned} {}^3\text{He} = & 0.8 \times \{0.01[2.8] (13.8[2.89] + 5.4[1.33] + 5[8.08] + 1.31[30.9] + 2[0.324]) \\ & + 0.01[10.7] (6.0[2.89] + 2.45[1.33] + 2.55[8.08] + 0.56[30.9] + 0.83[0.324]) \\ & + 0.4788 [2.8]\} \times 2.05 \times 10^{-4} / 9.79 \times 10^{-3} = 1.87 \times 10^{-1} \text{ atoms g}^{-1} \text{ yr}^{-1} \quad (14) \end{aligned}$$

Table 2. Composition of average crust.*

Element	Atomic Mass	Weight fraction element		
		Lower Crust	Middle Crust	Upper Crust
Li	6.94	6.00E-06	7.00E-06	2.00E-05
B	10.81	3.71E-06	3.22E-06	9.56E-06
C	12	5.88E-04	7.18E-03	3.24E-03
O	16	4.69E-01	4.87E-01	4.75E-01
Na	22.99	1.93E-02	2.37E-02	2.89E-02
Mg	24.31	4.28E-02	2.05E-02	1.33E-02
Al	26.92	8.78E-02	8.19E-02	8.04E-02
Si	28.09	2.44E-01	2.83E-01	3.09E-01
Cl	35.45	5.28E-05	5.28E-05	2.32E-04
K	39.1	4.98E-03	1.67E-02	2.82E-02
Ca	40.078	6.72E-02	3.64E-02	3.00E-02
Ti	47.87	4.79E-03	4.20E-03	3.00E-03
Cr	52	2.15E-04	8.30E-05	3.50E-05
Mn	54.94	7.74E-04	7.74E-04	6.20E-04
Fe	55.845	5.88E-02	4.48E-02	3.15E-02
Co	58.93	3.80E-05	2.50E-05	1.00E-05
Ni	58.69	8.80E-05	3.30E-05	2.00E-05
Sm	150.36	2.80E-06	4.40E-06	4.50E-06
Gd	157.25	3.10E-06	4.00E-06	3.80E-06
U	238.03	2.00E-07	1.60E-06	2.80E-06
Th	232.04	1.20E-06	6.10E-06	1.07E-05

*Rudnick and Fountain (1995); Parker (1967)

Wederpohl 1995; Gao et al. 1997; <http://earthref.org/GERM/>

Table 3. Neutron production, Li capture probability, and ${}^3\text{He}/{}^4\text{He}$ in average crust.

	Lower Crust	Middle Crust	Upper Crust
P_{th}	0.8	0.8	0.8
U(α ,n) neutrons/g/yr	2.51E-01	1.95E+00	3.58E+00
Th(α ,n) neutrons/g/yr	6.98E-01	3.42E+00	6.25E+00
Uf neutrons/g/yr	9.58E-02	7.66E-01	1.34E+00
$F_{6\text{Li}}$	6.99E-03	8.05E-03	2.09E-02
${}^3\text{He}$ atoms/g/yr	5.84E-03	3.95E-02	1.87E-01
${}^4\text{He}$ atoms/g/yr	1.57E+06	9.89E+06	1.73E+07
${}^3\text{He}/{}^4\text{He}$	3.71E-09	3.99E-09	1.08E-08

The ${}^3\text{He}$ and ${}^4\text{He}$ production rate for average lower, middle and upper crust compositions (Table 2) are shown in Table 3. It should be noted that the ${}^3\text{He}/{}^4\text{He}$ ratio is independent of [U] for constant U/Th.

Perhaps the most interesting observation from these calculations is the low value of ${}^3\text{He}/{}^4\text{He}$, even for the Li enriched upper crust. When we consider that the effect of radioelement heterogeneity (the preferential siting of U and Th in accessory phase minerals) would be to significantly reduce the neutron flux producing the ${}^3\text{He}$ (Figs. 2 and 3), these ${}^3\text{He}/{}^4\text{He}$ values must be considered to be upper limits for radiogenic production in average crust. Comparison with measured ${}^3\text{He}/{}^4\text{He}$ in whole rock samples, mineral separates and associated fluids is non-trivial. This is due to several factors:

(i) calibration and measurement of ${}^3\text{He}/{}^4\text{He}$ ratios $<10^{-8}$ is subject to significant error; (ii) sources of non-crustal ${}^3\text{He}$, such as mantle or cosmogenic ${}^3\text{He}$, can have a significant impact on the ${}^3\text{He}/{}^4\text{He}$ of a system; (iii) the siting and formation of ${}^3\text{He}$ and ${}^4\text{He}$ is different; and (iv) preferential release from different minerals or sites may lead to fractionation of the two isotopes.

A summary of measured and calculated values, both on whole

rocks and mineral separates is detailed in Mamyrin and Tolstikhin (1984) and more recently in Tolstikhin et al. (1996). These data compilations clearly show that a mantle ^3He signature (e.g., Graham 2002) can perturb the crustal $^3\text{He}/^4\text{He}$ even in ancient rocks (>2.8 Ga) by orders of magnitude. In samples where mantle and cosmogenic ^3He contributions to sedimentary rocks are discounted, variations from predicted ratios by a factor of three are ascribed to fractional release, with one exception, anhydrite, apparently requiring a fractionation factor of 20 (Tolstikhin et al. 1996).

Table 4. Selected measured and calculated $^3\text{He}/^4\text{He}$ in granites.

Sample ID	Location	Concentration (ppm)			$^3\text{He}/^4\text{He}$ Measured	$^3\text{He}/^4\text{He}$ Calc*	$^3\text{He}/^4\text{He}$ Calc**	Ref
		Li	U	Th				
WR	Rapakivi	36	6.7	35	1.60E-08	1.60E-08		1
Orthoclase	Rapakivi	6	2.5	2.7	4.00E-07			1
Biotite	Rapakivi	620	7.2	4.8	1.20E-07			1
Plagioclase	Rapakivi	10	5.9	5.3	5.00E-08			1
Quartz	Rapakivi	12	5.1	3.0	1.50E-07			1
Amphibole	Rapakivi	37	3.5	12.7	4.50E-08			1
Zircon	Rapakivi		990	150	2.00E-08			1
1-WR	Ukraine	37	4.8	44	1.60E-08	1.60E-08		2
2-WR	Ukraine	38	9.5	42	1.90E-08	1.70E-08		2
4-WR	Ukraine	26	11	85.4	3.00E-09	1.40E-08		2
43-WR	Tuva	10	1.6	10	8.00E-09	7.00E-09		2
27-WR	Caucasus	17	4.5	14	6.00E-08	9.00E-09		2
ss1-WR	Carmmenellis, UK	353	11.8	15.1	2.10E-09	8.00E-08	2.00E-08	3
ss1-fldspar	Carmmenellis, UK				2.38E-09			3
ss1-mica	Carmmenellis, UK				4.62E-09			3
ss1-quartz	Carmmenellis, UK				8.40E-10			3
rh11g-wr	Carmmenellis, UK	386	16.4	8.4	2.10E-09	6.40E-08	1.80E-08	3
rh12d-wr1	Carmmenellis, UK	300	10.2	6.6	1.54E-09	6.70E-08	1.80E-08	3
rh12dwr2	Carmmenellis, UK				4.90E-09	6.70E-08	1.80E-08	3
uraninite1	Carmmenellis, UK				9.80E-10			3
Water	Carmmenellis, UK				3.50E-08			3,4
WR	Stripa, Sweden	11	44.1	33		5.94E-09		5,6
Water	Stripa, Sweden				5.70E-09			5

* Assuming element homogeneity WR = Whole rock

** Assuming radioelements concentrated in uraninite grains with $\phi > 100\mu\text{m}$

1) Gerling et al. (1971) (in Mamyrin and Tolstikhin 1984)

2) Tolstikhin and Drubetskoy (1977) (in Mamyrin and Tolstikhin 1984)

3) Martel et al. (1990)

5) Andrews et al. (1989a)

4) Hilton et al. (1985)

6) Andrews et al. (1989b)

Granite systems provide an example where mantle influences appear to be minimal, and cosmogenic effects can be safely ruled out. This allows the effect of heterogeneity and release effects to be assessed. In Table 4 we have compiled selected $^3\text{He}/^4\text{He}$ data from different granite systems that also have theoretical values for comparison. With the exception of sample 27-WR, all measured whole rock values are either within error of calculated $^3\text{He}/^4\text{He}$ values or significantly lower. Two important observations can be made: i) more than one granite whole rock sample matches both measured and calculated $^3\text{He}/^4\text{He}$; and ii) the Stripa granite, which is the only system where the calculated neutron flux is confirmed with a measured neutron flux (Andrews et al. 1986), contains groundwater with $^3\text{He}/^4\text{He}$ indistinguishable from predicted production (Andrews et al. 1989a;b). Although a small data set, an important inference is that many systems display no resolvable fractionation from theoretical values.

Where fractionation is observed, the ‘nugget’ effect must be first ruled out. This is when a whole rock sample contains an accessory mineral concentration in excess of the average rock. Accessory phases, such as uraninite, contain high concentrations of He with low $^3\text{He}/^4\text{He}$ (Table 4) and will result in a $^3\text{He}/^4\text{He}$ lower than the average bulk rock. In the case of the Carnmenellis granite (Martel et al. 1990) this cause can be ruled out by multiple whole rock analyses giving the same $^3\text{He}/^4\text{He}$ value. If the measured uraninite size of $\sim 50\text{-}\mu\text{m}$ radius is representative of the radioelement distribution, this rock would be expected to produce $^3\text{He}/^4\text{He} = 1 \times 10^{-8}$. Nevertheless, measured whole rock and mineral separate analyses are a factor of three too low to be accounted for by radioelement heterogeneity reduced flux alone (Table 4). An additional factor of three to five is required and is probably release related (Martel et al. 1990). A complementary ^3He -enriched fluid would be expected to be observed. This is indeed seen in the associated groundwater (Table 4). It would appear that the order of magnitude difference between the whole rock and groundwater $^3\text{He}/^4\text{He}$ in the Carnmenellis granite can be ascribed to a combination of radioelement heterogeneity and preferential release of ^3He into the surrounding fluid system. If however, the average uraninite grain size is an order of magnitude smaller, all of the difference between whole rock and groundwater $^3\text{He}/^4\text{He}$ has to be ascribed to preferential ^3He release into the surrounding groundwater.

In summary, the effect of radioelement heterogeneity on $^3\text{He}/^4\text{He}$ in most systems is probably small. There is some evidence from the Carnmenellis system that fractionation of $^3\text{He}/^4\text{He}$ due to preferential ^3He release could be as high as a factor of 10, but taking into account the evidence for radioelement element heterogeneity this is more reasonably, at most, a factor of 3 (e.g., Tolstikhin et al. 1996). In large crustal fluid systems reasonably sampling ‘average’ crust, $^3\text{He}/^4\text{He}$ ratios in excess of $1\text{--}3 \times 10^{-8}$ (i.e., more than three times the upper crust value) are due to a resolvable ^3He excess from sources external to the crust (e.g., Marty et al 1993).

Neon

The production of Ne isotopes in the crust is entirely due to nucleogenic routes. Recognized by (Wetherill 1954), the only significant production routes are

$^{17,18}\text{O}(\alpha, n)^{20,21}\text{Ne}$, $^{19}\text{F}(\alpha, n)^{22}\text{Na}(\beta^+)^{22}\text{Ne}$, $^{24,25}\text{Mg}(n, \alpha)^{21,22}\text{Ne}$, $^{23}\text{Na}(n, \alpha)^{20}\text{Ne}$, $^{19}\text{F}(\alpha, p)^{22}\text{Ne}$ (Yatsevich and Honda 1997). Their rate of production is therefore related to radioelement and target-element concentrations as well as the distribution of the target element with respect to any radioelement heterogeneity.

The first investigations of Ne production rates in the crust include the work by Sharif-Zade et al. (1972), Shukolyukov et al. (1973) and Verkhovskiy and Shukolyukov (1976a,b). In addition to (α, n) production, Rison (1980) also investigated (n, α) production rates. This work however, used oxygen (α, n) yields from Feige (1968) which results in a $^{20,21}\text{Ne}$ yield twice as high as more recent measurements of the (α, n) yields by West and Sherwood (1982). More recently it has also been shown by Yatsevich and Honda (1997) that the neutron yields from the (n, α) routes calculated by Rison (1980) are also too high by a factor of five because of an overestimation of the neutron energy spectrum (Fig. 4). Using the revised (α, n) yields from West and Sherwood (1982), Hünemohr (1989) and Yatsevich and Honda (1997) have calculated the Ne (α, n) yields for average crust and mantle materials. Using only the reaction cross sections Leya and Wieler (1999) have also estimated Ne production in average crust. These production rates are summarized in Table 5. Yatsevich and Honda (1997) show that with the updated Mg (n, α) yields, production by this route accounts for $<0.13\%$ of the ^{21}Ne and ^{22}Ne produced in the crust. In the mantle this production mechanism is more important, with some 3.57% and 65.35% of ^{21}Ne and ^{22}Ne produced by Mg (n, α) reactions, respectively.

Table 5. Comparison of Nucleogenic Ne Production Calculations.

Isotope	Reference	Production over 4.5 Ga		Present-day production rate	
		cm ³ STP/g		cm ³ STP/g year	
		Mantle	Crust	Mantle	Crust
²⁰ Ne	Leya and Wieler, 1999	1.10E-13	1.16E-11	1.42E-23	1.49E-21
	Yatsevich and Honda, 1997	1.35E-13	1.45E-11	1.74E-23	1.86E-21
	Hunemohr, 1989	1.45E-13	1.54E-11	1.87E-23	1.98E-21
	Kyser and Rison, 1982	2.50E-13	2.70E-11	3.40E-23	3.70E-21
²¹ Ne	Leya and Wieler, 1999	1.15E-12	1.22E-10	1.48E-22	1.56E-20
	Yatsevich and Honda, 1997	1.45E-12	1.50E-10	1.86E-22	1.92E-20
	Hunemohr, 1989	1.50E-12	1.59E-10	1.92E-22	2.04E-20
	Kyser and Rison, 1982	3.20E-12	3.00E-10	4.20E-22	4.00E-20
²² Ne	Leya and Wieler, 1999	1.13E-14	4.05E-11	1.48E-24	5.23E-21
	Yatsevich and Honda, 1997	3.02E-14	4.10E-11	3.98E-24	5.23E-21
	Hunemohr, 1989	1.17E-14	4.58E-11	1.50E-24	5.87E-21
	Kyser and Rison, 1982	1.00E-13	7.50E-13	1.30E-23	1.20E-22
⁴ He	Equation (13)			4.13E-15	4.13E-13

Although Hünemohr (1989) and Leya and Wieler (1999) do not consider neutron production routes, because these reactions are insignificant in the crust, this enables a direct assessment of the different Ne production rate estimates. The values given by Hünemohr are ~7% higher than those by Yatsevich and Honda, while the values of Leya and Wieler are ~25% lower. The marked difference between the approach of Leya and Wieler (1999) and the other workers was in the adoption of cross section data rather than thick target yields and is independent of limiting assumptions such as stopping ratios. Nevertheless, this comparison provides the level of uncertainty that should be used when applying these theoretical production rates.

Empirical production rates for variable U, Th and rock composition can be derived from the absolute rates for the different reactions in Yatsevich and Honda (1997) and the proportion ascribed to U relative to Th in Hünemohr (1989, p105) to give:

$$^{20}\text{Ne} = \frac{1.74 \times 10^{-23}}{0.02} \times \left(\frac{4.48}{6.1} [\text{U}] + \frac{1.62}{6.1} \frac{[\text{Th}]}{3} \right) \left(0.9978 \frac{[\text{O}]}{44} + 0.0022 \frac{[\text{Na}]}{1} \right) \\ = (6.39[\text{U}] + 0.770[\text{Th}])(0.0226[\text{O}] + 0.0022[\text{Na}]) \times 10^{-22} \quad (15)$$

$$^{21}\text{Ne} = \frac{1.86 \times 10^{-22}}{0.02} \times \left(0.9646 \frac{[\text{O}]}{44} \left(\frac{45}{62} [\text{U}] + \frac{17}{62} \frac{[\text{Th}]}{3} \right) + \right. \\ \left. 0.0357 \frac{[\text{Mg}]}{21} \left(\frac{11.7}{17.7} [\text{U}] + \frac{6.0}{17.7} \frac{[\text{Th}]}{3} \right) \right) \\ = \{(1.48[\text{U}] + 0.186[\text{Th}])[\text{O}] + (0.105[\text{U}] + 0.0179[\text{Th}])[\text{Mg}]\} \times 10^{-22} \quad (16)$$

$$^{22}\text{Ne} = \frac{3.98 \times 10^{-24}}{0.02} \times \left(\begin{aligned} &0.3465 \frac{[\text{F}]}{16} \left(\frac{0.340}{0.479} [\text{U}] + \frac{0.139}{0.479} \frac{[\text{Th}]}{3} \right) \\ &+ 0.6535 \frac{[\text{Mg}]}{21} \left(\frac{2.75}{4.05} [\text{U}] + \frac{1.30}{4.05} \frac{[\text{Th}]}{3} \right) \end{aligned} \right)$$

$$= \{ (3.06[\text{U}] + 0.417[\text{Th}])[\text{F}] + (4.20[\text{U}] + 0.663[\text{Th}])[\text{Mg}] \} \times 10^{-24} \quad (17)$$

Production rates in $\text{cm}^3 \text{ STP g}^{-1} \text{ yr}^{-1}$; U, Th and F concentrations in ppm, O, Mg and Na in wt %. Initial constants have been reduced as little as possible for the purpose of transparency with the original references. It should be noted that these equations neglect the effect of variable stopping power and should be treated with caution when applied to systems with an average atomic mass significantly different from average mantle or crust. Comparison of relative Ne production rates will be less sensitive to density variables. In addition, Leya and Wieler (1999) have shown that if you lower the O-concentration then you typically increase the stopping power and increase the production within this mineral. The net result is that the $^4\text{He}/^{21}\text{Ne}$ production ratios show no strong dependence on the target chemistry because both effects typically cancel out.

Hünemohr (1989), and later Eikenberg et al. (1993), have measured both the elemental composition of various accessory phase minerals and the nucleogenic ^{21}Ne and ^{22}Ne concentration trapped within the minerals (Table 6). Because of the small contribution from Mg in these minerals the predicted $^{21}\text{Ne}/^{22}\text{Ne}$ is shown to be roughly proportional to the O/F elemental ratio. A comparison of $^{21}\text{Ne}/^{22}\text{Ne}$ calculated from the measured O and F composition (Table 6) and using Equations (16) and (17) essentially reproduces the observed $^{21}\text{Ne}/^{22}\text{Ne}$ in these minerals (Fig. 5). Clearly on a mineral scale the relative rates of ^{21}Ne , ^{22}Ne and therefore ^{20}Ne production are well constrained theoretically and confirmed experimentally.

Ne isotopes in crustal fluids. The Ne isotope data recorded in natural gases and brines, by mass balance, reasonably preserve a record of regional crustal systems. The first studies of Ne isotopes in natural gases observed $^{21}\text{Ne}/^{22}\text{Ne}$ ratios below those found in local radioactive minerals. This was interpreted to be due to preferential thermal release of ^{22}Ne from the producing mineral (Shukolyukov et al. 1973). More recently Kennedy et al. (1990) showed that Ne isotopes in natural gases defined a crustal source with $^{21}\text{Ne}/^{22}\text{Ne} = 0.469 \pm 0.006$, consistent with an O/F ratio of 113 (Fig. 6). This is much lower than average crustal values of O/F = 752 or crystalline shield values of O/F = 1140, that would produce crustal $^{21}\text{Ne}/^{22}\text{Ne}$ values of 3.51 and 5.18 respectively. After assessing the potential of either regionally low O/F ratios or an underestimation of ^{22}Ne production rates, Kennedy et al. (1990) concluded that these values in fact reflect the average O/F ratio of the mineral environment in which the U and Th are sited, rather than average crustal values. The uniformity of the $^{21}\text{Ne}/^{22}\text{Ne}$ production ratio from a variety of locations lead these workers to further conclude that the source of the ‘unusual’ O/F ratios necessitated a common and widely distributed mineral suite such as micas and amphiboles. It was argued that other U-Th-F-rich minerals such as apatite and fluorite are more commonly found in vein deposits, and thus more likely to be inhomogeneously distributed in the crust.

A data inversion based on He and Ne isotopic mixing between crust and magmatic end-members was used to resolve an independent crustal $^{21}\text{Ne}/^{22}\text{Ne}$ production ratio of 0.57 ± 0.1 in a geographically more diverse sample suite (Ballentine 1997). These natural gases, from a variety of extensional environments, are assumed to mix with a sub-continental lithospheric mantle component with $^3\text{He}/^4\text{He}$ of 6.0 Ra (where Ra is the atmospheric $^3\text{He}/^4\text{He}$ ratio). The use of isotopes alone removes the possibility of ele-

mental fractionation effects once the mantle fluids have mixed with the crustal gases. The resolved crustal $^{21}\text{Ne}/^{22}\text{Ne}$ production ratio lends further support to the uniformity of the mineral scale relationship between O, F and U+Th in the crust established by Kennedy et al (1990).

Table 6. Comparison of measured and calculated $^{21}\text{Ne}/^{22}\text{Ne}$ production in accessory phase minerals.

<i>Sample</i>	<i>U ppm</i>	<i>Th ppm</i>	<i>U/Th</i>	<i>O %</i>	<i>Mg %</i>	<i>F ppm</i>
Bastnäsite	150	2600	0.0577	26.6	0	73500
Thorite	600	283000	0.0021	29.8	0	18600
Fergusonite	150	600	0.2500	40.4	0.09	14200
Xenotime	1000	16000	0.0625	46.5	1.58	2980
Priorite	72000	9500	7.58	27.5	0	1360
Euxenite	58000	16000	3.63	31.2	0	1100
Euxenite	41500	12000	3.46	31.3	0	910
Thorite	10000	174000	0.0575	48.9	0.11	540
Monazite	1400	32000	0.0438	27.7	0	340
Gololinite	900	2300	0.3913	35.5	0.013	210
Samarskite	63000	7400	8.51	23.4	0	180
Monazite	800	40000	0.0200	29.2	0.01	90

<i>Sample</i>	$^{21}\text{Ne}/^{22}\text{Ne}$ <i>Calc</i>	$^{21}\text{Ne}/^{22}\text{Ne}$ <i>Meas</i>	<i>error</i>	<i>Meas/Calc</i>
Bastnäsite	0.0171	0.0091	0.0025	0.53
Thorite	0.0758	0.1039	0.0266	1.37
Fergusonite	0.1346	0.1013	0.0265	0.75
Xenotime	0.7395	0.81	0.23	1.10
Priorite	0.9566	0.88	0.23	0.92
Euxenite	1.3418	0.82	0.21	0.61
Euxenite	1.6272	1.00	0.26	0.61
Thorite	4.2835	6.56	1.99	1.53
Monazite	3.8542	5.05	1.33	1.31
Gololinite	7.9969	5.2	1.4	0.65
Samarskite	6.1501	13.0	3.4	2.11
Monazite	15.347	7.7	2.3	0.50
Average				1.00
				± 0.1433

Composition and $^{21}\text{Ne}/^{22}\text{Ne}$ analysis of U- and Th-rich minerals by Hühemohr (1989).

Calculated $^{21}\text{Ne}/^{22}\text{Ne}$ using Equations (16) and (17).

Because of the incompatibility of F, the siting of this element in the crust has been assumed to be in accessory phase minerals. The close relationship between F and U+Th suggests that a significant portion of the crust U+Th budget must also be sited in the same accessory phase minerals. Neither Kennedy et al. (1990) or Ballentine (1997) considered preferential ^{22}Ne release. However, $^4\text{He}/^{40}\text{Ar}$ ratios are sensitive to thermal fractionation and release effects (e.g., Ballentine et al. 1994). Natural gases that show no fractionation from predicted crustal $^4\text{He}/^{40}\text{Ar}$ ratios and relatively uniform crustal $^4\text{He}/^{21}\text{Ne}$ still show crustal $^{21}\text{Ne}/^{22}\text{Ne}$ production rates below those predicted for average crust (see Fig. 7 for sample data sets). $^{21}\text{Ne}/^{22}\text{Ne}$ production below average crust is probably not due to excess ^{22}Ne from release fractionation effects, but is due to radioelement siting and target element heterogeneity in the crust.

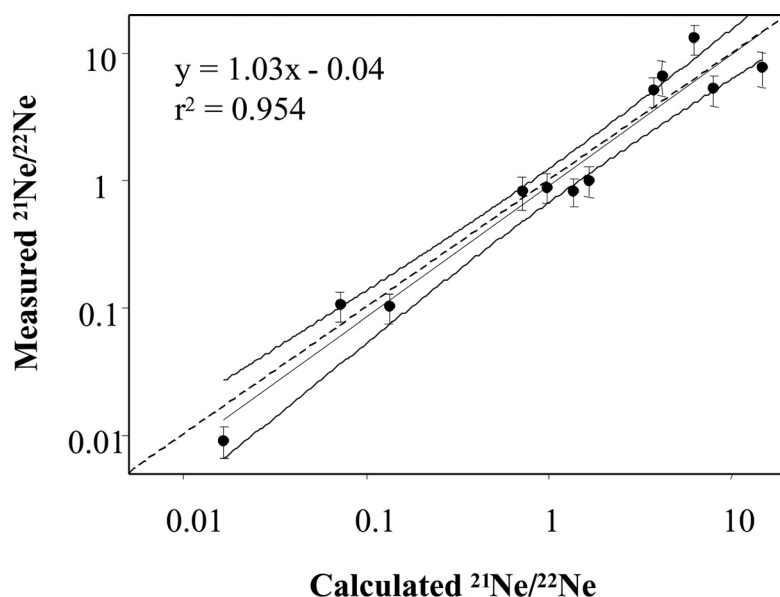
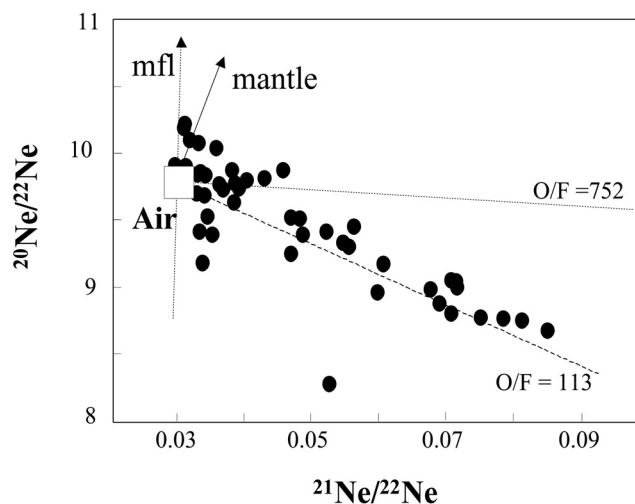


Figure 5. Comparison of values measured by Hünemohr (1989) in U- and Th-rich minerals and those calculated using the Equations (16) and (17). The outer lines show the 95% confidence limit of the fit. The dashed line is the ideal fit with a gradient of 1.

Figure 6. $^{20}\text{Ne}/^{22}\text{Ne}$ vs $^{21}\text{Ne}/^{22}\text{Ne}$ in crustal fluids after Kennedy et al. (1990) showing significant deviation from the crustal $^{21}\text{Ne}/^{22}\text{Ne}$ mixing line predicted for an average crustal O/F ratio of 752. Data from the Alberta Basin and New Mexico (Kennedy et al. 1990), Brines from the Canadian shield (Bottomley et al. 1984) and helium-rich gases from the SW USA (Emerson et al. 1966). The two arrows show vectors predicted for mass fractionation (mfl) and mantle fluid admixtures (mantle) that probably account for the scatter observed.



Several workers have noted the possible underestimate of ^{22}Ne production in the mantle because of similar accessory phase associations of U+Th with F (Barfod 1999; Niedermann et al. 1997). Taking the average of the two crustal studies discussed above and an error that encompasses both values in a T-test, we recommend for average crustal calculations a $^{21}\text{Ne}/^{22}\text{Ne}$ crustal production ratio of 0.52 ± 0.04 . ^{20}Ne production is dominated, like ^{21}Ne , by reactions with oxygen. The $^{21}\text{Ne}/^{20}\text{Ne}$ production ratios will not therefore be affected by element heterogeneity.

Observed $^4\text{He}/^{21}\text{Ne}$. An important question is whether the observed production rate of Ne isotopes, relative to ^4He , is also in good agreement with the theoretical calculations. Since ^{21}Ne is largely dependent on the oxygen concentration, which is ubiquitous in the crust, the $^4\text{He}/^{21}\text{Ne}$ ratio should be the least affected by element heterogeneity. Hünemohr (1989) measured the $^4\text{He}/^{21}\text{Ne}$ ratio produced in the same mineral suite as the Ne-isotope study discussed above. Results from these measurements for individual minerals ranged from predicted production values to $^4\text{He}/^{21}\text{Ne}$ values an order of magnitude lower. This is attributed to variable but preferential ^4He loss from some samples.

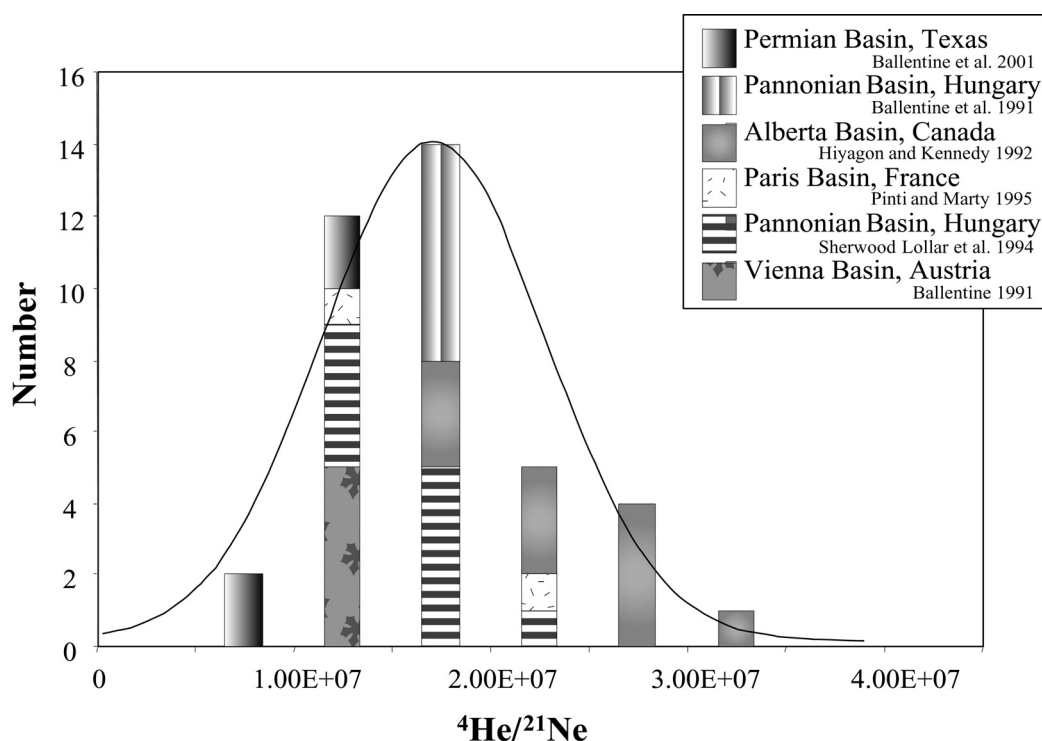


Figure 7. Observed $^4\text{He}/^{21}\text{Ne}^*$ production ratio in the crust. We have taken all natural gas data available with $^4\text{He}/^{40}\text{Ar}^*$ ratios in the range 3-6 and with $^3\text{He}/^4\text{He} < 0.6\text{Ra}$ to remove bias due to elemental fractionation or mantle components. The mean from 38 samples is $1.71 \pm 0.09 \times 10^7$ (1 standard error). This compares with theoretical estimates of between 2.02 - 2.64×10^7 (Table 5).

Another approach is to investigate the noble gas composition of natural gases and other crustal fluids that have collected noble gases from large crustal volumes. Many samples however, show significant elemental fractionation (e.g., Ballentine et al. 1991). Fractionation may occur by preferential release of noble gases from the producing minerals, transport effects such as diffusion, or water-oil-gas phase equilibrium fractionation. Although these processes have little effect on isotopic ratios, they provide the limiting factor in a comparison of elemental ratios and are not considered in earlier estimates of measured $^4\text{He}/^{21}\text{Ne}$ (Verchovskiy et al. 1977; Tolstikhin 1978; Kennedy et al. 1990).

Other noble gases however, provide a sensitive indicator of fractionation. Making the assumption that the present day average crustal production $^4\text{He}/^{40}\text{Ar}$ ratio is ~ 5 (next section), we have compiled in Figure 7 all the available data from regional crustal fluid systems for which crustal $^4\text{He}/^{40}\text{Ar} = 5.0 \pm 1.0$. We have also excluded samples with magmatic He contributions greater than $\sim 10\%$ ($^3\text{He}/^4\text{He} > 0.6\text{Ra}$). Data without full He, Ne and Ar abundance and isotopic information are not included. No correlation is observed between $^4\text{He}/^{40}\text{Ar}$ and $^4\text{He}/^{21}\text{Ne}$ in this filtered data set. The data distribution approximates a Gaussian distribution with a mean observed crustal $^4\text{He}/^{21}\text{Ne}$ production rate of $1.71 \pm 0.09 \times 10^7$. This compares with the theoretical estimates for average crust of between 2.02 - 2.64×10^7 (Table 5). At present it is unclear why the crustal record preserves a ^{21}Ne production rate 15-35% greater than that the theoretical calculations would predict.

Argon

^{40}Ar production in the crust is dominated by the decay of ^{40}K , and is therefore directly proportional to the K concentration. ^{40}K has a branched decay mode, producing ^{40}Ca by beta decay and ^{40}Ar by electron capture, with decay constants of $\lambda_\beta =$

$4.962 \times 10^{-10} \text{ yr}^{-1}$ and $\lambda_e = 0.581 \times 10^{-10} \text{ yr}^{-1}$ respectively (Steiger and Jäger 1977). The total decay constant of ^{40}K is therefore given by:

$$\lambda_K = \lambda_\beta + \lambda_e = 5.543 \times 10^{-10} \text{ yr}^{-1} \quad (18)$$

(It should be noted that this value has recently been re-determined to give $\lambda_K = (5.463 \pm 0.054) \times 10^{-10} \text{ yr}^{-1}$, although usage of this newer value is not yet widespread (Begemann et al. 2001; Kelley 2002)). The fraction of ^{40}K that decays to ^{40}Ar is therefore λ_e/λ_K and the ^{40}Ar production rate can be expressed as a function of K (in ppm), [K], by:

$$^{40}\text{Ar atoms g}^{-1} = X_K [\text{K}] \times 10^{-6} (N_A/A_K) (\lambda_e/\lambda_K) (e^{\lambda t} - 1) \quad (19)$$

where X_K = fractional natural abundance of $^{40}\text{K} = 1.167 \times 10^{-4}$, N_A = Avogadro's number (6.023×10^{23}), A_K = molar mass of K = 39.964g, and t = age (yr). The present day production can therefore be expressed as:

$$^{40}\text{Ar atoms g}^{-1} \text{ yr}^{-1} = 102.2 [\text{K}] \quad (20)$$

Combining Equation (20) with Equation (13), the term for He production in the crust, the $^4\text{He}/^{40}\text{Ar}$ ratio in the crust is defined by the (U+Th)/K ratio, where:

$$^4\text{He}/^{40}\text{Ar} = \{(3.115 \times 10^6 + 1.272 \times 10^5) [\text{U}] + 7.710 \times 10^5 [\text{Th}]\} / 102.2 [\text{K}] \quad (21)$$

Using the crustal compositions from Table 2, Equation (21) predicts present day $^4\text{He}/^{40}\text{Ar}$ production ratios in the lower, middle and upper crust of 3.09, 5.79 and 6.0 respectively, with a production weighted average (Table 2) of 5.7. In principle measured crustal $^4\text{He}/^{40}\text{Ar}$ ratios can provide a test of these K/U ratios (Dymond and Hogan 1973), an issue that has recently provided significant controversy (e.g., Albarède 1998). In practice $^4\text{He}/^{40}\text{Ar}$ ratios are sensitive to fractionation during thermal release from their respective mineral sites (Ballentine et al. 1994; Mamyrin and Tolstikhin 1984), as well as subsequent transport and phase related fractionation processes (Ballentine et al. 1991). We have adopted in the Ne section above the assumption that $^4\text{He}/^{40}\text{Ar}$ ratios within the range 4 to 6 are relatively unaffected by these process to filter out fractionated samples, but with the implicit assumption that the K/U ratio of the crust is well defined. We discuss later the information about release, transport and phase fractionation available by considering deviations from predicted crustal values. Also see Ballentine et al. (2002).

^{36}Ar production in the crust is small compared to the ambient background of atmosphere-derived ^{36}Ar introduced into the crust dissolved in groundwater and is usually neglected. Although a small amount of muon-induced ^{36}Ar occurs close to the surface, the principle route of production is the β -decay of ^{36}Cl (Fontes et al. 1991; Hünemohr 1989). ^{36}Cl has a half-life of $3.01 \times 10^5 \text{ yr}$ and decays to ^{36}Ar with a branching ratio, R , of 0.95. ^{36}Cl is only produced in the crust by the thermal neutron reaction $^{35}\text{Cl}(n,\gamma)^{36}\text{Cl}$ (Bentley et al. 1986; Fontes et al. 1991). The fraction of thermal neutrons captured by any one element, F_i , is given by Equation (11). It should be noted that the reaction cross section given for Cl in Table 1 is the combined probability for ^{35}Cl and ^{37}Cl with thermal neutron capture cross sections and relative abundances of 43 and 0.43 barns and 75.77% and 24.23% respectively. These values give $F_{^{35}\text{Cl}} = 0.02206$ for average upper crust. For a system in steady state the number of ^{36}Cl atoms present is given by:

$$^{36}\text{Cl atoms g}^{-1} = P_{\text{th}} N F_{^{35}\text{Cl}} / \lambda_{^{36}\text{Cl}} \quad (22)$$

where P_{th} is the probability of a neutron reaching thermal energy, N is the neutron density given by Equation (10) and $\lambda_{^{36}\text{Cl}}$ the decay constant of ^{36}Cl . Once this equilibrium has been attained the production of ^{36}Cl is equal to its rate of decay to ^{36}Ar , giving the following ^{36}Ar production rate:

$$^{36}\text{Ar atoms g}^{-1} \text{ yr}^{-1} = P_{\text{th}} N R F_{35\text{Cl}} \quad (23)$$

For average upper crust (Table 2), $N = 10.6$ neutrons $\text{g}^{-1} \text{ yr}^{-1}$ (Table 3), giving a ^{36}Ar production rate of 0.19 atoms $\text{g}^{-1} \text{ yr}^{-1}$. This compares with ^{40}Ar production in the upper crust of 2.93×10^6 atoms $\text{g}^{-1} \text{ yr}^{-1}$, to give a crustal $^{40}\text{Ar}/^{36}\text{Ar}$ production ratio of 1.54×10^7 . Fontes et al (1991) discuss how ^{36}Ar rates can be a significant factor in specific Cl-U-Th-rich environments and further discuss cosmogenic rates of ^{36}Ar production.

^{38}Ar excess relative to ^{36}Ar observed in early studies of U- and Th-rich minerals led workers to investigate the $^{35}\text{Cl}(\alpha, p)^{38}\text{Ar}$ and $^{41}\text{K}(n, \alpha)^{38}\text{Ar}$ production routes of ^{38}Ar (Fleming 1953; Wetherill 1954). Hünemohr (1989) and Eikenberg et al. (1993) in addition investigated the production via $^{37}\text{Cl}(n, \gamma)^{38}\text{Cl}(\beta^-)^{38}\text{Ar}$. $^{41}\text{K}(n, \alpha)^{38}\text{Ar}$ is energetically un-favorable and can be discounted. Although energetically favorable, the thermal neutron interaction with ^{37}Cl has a reaction cross-section of only 0.43 barns and a natural abundance of 24.23%. The ^{38}Ar production rate from thermal neutrons is given by:

$$^{38}\text{Ar atoms g}^{-1} \text{ yr}^{-1} = P_{\text{th}} N F_{37\text{Cl}} \quad (24)$$

For average upper crust (Table 2) $F_{37\text{Cl}} = 6.7 \times 10^{-5}$ to give a ^{38}Ar production rate of 6×10^{-4} atoms $\text{g}^{-1} \text{ yr}^{-1}$. No modeling of the $^{35}\text{Cl}(\alpha, p)^{38}\text{Ar}$ production rates have been made, although the ratio of the nuclear cross sections for $^{35}\text{Cl}(\alpha, p)^{38}\text{Ar}$ and $^{19}\text{F}(\alpha, n)^{22}\text{Ne}$ is predicted from Woosley et al. (1975) to be ~ 0.2 for α -energies between 4 and 8 MeV (Eikenberg et al. 1993). Both Hünemohr (1989) and Eikenberg et al. (1993) show that $^{38}\text{Ar}/^{22}\text{Ne}$ excesses are correlated with $^{35}\text{Cl}/^{19}\text{F}$ ratios in (U+Th)-rich minerals with a gradient consistent with this estimate. The relationship between relative F and Cl α -reaction cross sections enables an empirical equation for ^{38}Ar production rates to be derived from Equation (17):

$$\begin{aligned} ^{38}\text{Ar} &= \frac{3.98 \times 10^{-24}}{0.02} \times 0.3465 \times 0.2423 \times 0.2 \times \frac{19}{37} \times \frac{[\text{Cl}]}{16} \left(\frac{0.340}{0.479} [\text{U}] + \frac{0.139}{0.479} \frac{[\text{Th}]}{3} \right) \\ &= \{[\text{Cl}](0.76[\text{U}] + 0.104[\text{Th}])\} \times 10^{-25} \end{aligned} \quad (25)$$

[Cl], [U] and [Th] are concentrations in ppm and the production rate of ^{38}Ar in $\text{cm}^3 \text{ STP g}^{-1} \text{ yr}^{-1}$. For average upper crust (Table 2) this gives a production rate of 0.002 atoms $\text{g}^{-1} \text{ yr}^{-1}$. Inclusion of $^{37}\text{Cl}(n, \gamma)$ derived ^{38}Ar results in a total 0.0026 atoms $\text{g}^{-1} \text{ yr}^{-1}$, to give a $^{38}\text{Ar}/^{36}\text{Ar}$ crustal production ratio of ~ 0.014 . This value is calculated assuming an homogenous distribution of elements in the upper crust. Unlike ^{36}Ar , the ^{38}Ar production is dominated by α -particle reactions and the production rate of ^{38}Ar will be sensitive to element heterogeneity and the siting of Cl relative to U+Th in the crust. This is directly analogous to ^{22}Ne production rates and is illustrated in Cl and (U+Th)-rich minerals by observed $^{38}\text{Ar}/^{36}\text{Ar}$ production ratios in excess of 14.7 (Eikenberg et al. 1993), some 10^3 times higher than ratios predicted for an elementally homogenous crust. Although some natural gases show $^{38}\text{Ar}/^{36}\text{Ar}$ ratios in excess of air ratios (Ballentine 1991), pointing to average crustal production ratios in excess of calculated average values, no systematic assessment of the ^{38}Ar production rate in the crust exists. Estimates of nucleogenic contributions to the atmosphere based on average crustal production rates will underestimate the nucleogenic ^{38}Ar contribution, while the effect of element heterogeneity on $^{38}\text{Ar}/^{36}\text{Ar}$ ratios in the mantle has yet to be assessed.

Small amounts of the unstable gases ^{37}Ar and ^{39}Ar are produced in the crust through the thermal neutron reactions $^{39}\text{K}(n, p)^{39}\text{Ar}$ and $^{40}\text{Ca}(n, \alpha)^{37}\text{Ar}$ (Lehmann et al. 1993; Pearson Jr. et al. 1991). ^{37}Ar and ^{39}Ar have half-lives of 34.95 ± 0.08 days (Renne and Norman 2001) and 269 years respectively. Applications of these tracers are reviewed by Kipfer et al. (2002, this volume).

Krypton and xenon

Fission of ^{238}U provides the dominant mechanism for the production of $^{83,84,86}\text{Kr}$ and $^{129,131,132,134,136}\text{Xe}$ in the crust today, while production of the shielded isotopes $^{80,82}\text{Kr}$ and $^{124,126,128,130}\text{Xe}$ can be neglected. Contributions from the spontaneous fission of ^{232}Th and thermal or fast neutron induced fission of ^{235}U , ^{238}U and ^{232}Th , must also be taken into consideration. Although there has been considerable past interest in the determination of the various fission spectra and yields, recent investigation of the U-Xe-Kr dating tool pioneered by Shukolyukov et al. (1974), has resulted in a re-determination of many of these values (Eikenberg et al. 1993; Ragettli et al. 1994; Wieler and Eikenberg 1999) (Table 7). Fission products of ^{244}Pu and the decay products of ^{129}I , both now extinct, contribute important Kr and Xe isotopic components to both the terrestrial mantle and atmosphere (Porcelli and Ballentine 2002). Kr and Xe derived from these extinct radionuclides are not produced in the crust hence do not contribute to the crustal system except where carried in as components of magmatic or atmosphere-derived fluids.

It is convenient to consider the production rate of ^{136}Xe by the various routes, and scale the yield of the other Kr and Xe fissiogenic isotopes to this product (Table 7). ^{136}Xe production by spontaneous fission in the crust is directly proportional to the ^{238}U concentration. ^{238}U has a branched decay mode, producing ^{234}Th by α -decay with $\lambda_{\alpha} = 1.55 \times 10^{-10} \text{ yr}^{-1}$ (Steiger and Jäger 1977), and spontaneous fission producing amongst other isotopes $^{136}\text{Xe}_{\text{sf}}$ with $\lambda_{\text{sf}} \sim 9 \times 10^{-17} \text{ yr}^{-1}$. Because $\lambda_{\text{sf}} \ll \lambda_{\alpha}$ $\square$ $\lambda_{238\text{U}} \approx \lambda_{\alpha}$, and the production rate of $^{136}\text{Xe}_{\text{sf}}$ is given by:

$$^{136}\text{Xe}_{\text{sf}} \text{ atoms g}^{-1} \text{ yr}^{-1} = (N_{\text{A}}/A_{238\text{U}}) \times 10^{-6} X_{238} [\text{U}] (^{136}\text{Y} \lambda_{\text{sf}} / \lambda_{\alpha}) (e^{\lambda_{\alpha} t} - 1) \quad (26)$$

where $[\text{U}]$ is the concentration of U in ppm, ^{136}Y is the yield (~ 0.063), or fraction of the fission product resulting in ^{136}Xe production, X_{238} is the fractional natural abundance of ^{238}U (0.992745), N_{A} is Avogadro's number (6.023×10^{23}), A_{238} is the molar mass of ^{238}U (238.051g), and t = age (yr). Although the values for ^{136}Y and λ_{sf} have been independently determined, it is more convenient and precise to determine their product, $^{136}\text{Y} \lambda_{\text{sf}}$. $^{136}\text{Y} \lambda_{\text{sf}}$ has been determined in a variety of minerals and shows a bimodal distribution of about 20% variation between U-oxides and U-rich accessory phase minerals (Eikenberg et al. 1993; Ragettli et al. 1994). Ragettli et al. (1994) argue that the lower values in U-oxides can be attributed to several possible causes, including changes in oxidation state, recoil loss and diffusion. In contrast, through a careful study Ragettli et al. (1994) show that the higher values recorded in the accessory phase minerals cannot be attributed to either underestimation of mineral formation ages, overestimation of recoil effects, excess parent-less Xe or fissiogenic ^{136}Xe from other sources within the minerals. Ragettli et al. (1994) give a best estimate for $^{136}\text{Y} \lambda_{\text{sf}} = 6.83 \pm 0.18 \times 10^{-18} \text{ yr}^{-1}$. For the average upper crust U concentrations (Table 2), this gives a $^{136}\text{Xe}_{\text{sf}}$ production rate of $0.0480 \text{ atoms g}^{-1} \text{ yr}^{-1}$.

Estimates of λ_{sf} for ^{232}Th are at least two orders of magnitude smaller than λ_{sf} for ^{238}U , between 5.4×10^{-19} to $< 2.1 \times 10^{-21} \text{ yr}^{-1}$ (Segrè 1952; Wieler and Eikenberg 1999), and this source of ^{136}Xe production can be neglected. Experimental work shows that in pitchblendes and uraninites, no fast neutron induced fission component from ^{238}U can be resolved, and that all Kr and Xe isotope data can be accounted for with only three components: air; ^{238}U spontaneous fission products, and thermal neutron induced fission of ^{235}U (Fig. 8) (Eikenberg et al. 1993). The rate of thermal neutron fission production of $^{136}\text{Xe}_{\text{nf}}$ can be calculated using the following:

$$^{136}\text{Xe}_{\text{nf}} \text{ atoms g}^{-1} \text{ yr}^{-1} = P_{\text{th}} N F_{235\text{U}} ^{136}\text{Y}_{\text{nf}} \quad (27)$$

Table 7. Xe and Kr Fission Spectra and Yields.

	$^{83}\text{Kr}/^{86}\text{Kr}$	$^{84}\text{Kr}/^{86}\text{Kr}$	$^{129}\text{Xe}/^{136}\text{Xe}$	$^{131}\text{Xe}/^{136}\text{Xe}$	$^{132}\text{Xe}/^{136}\text{Xe}$	$^{134}\text{Xe}/^{136}\text{Xe}$	$^{136}\text{Xe}/^{86}\text{Kr}$	$^{136}\text{Xe yield}$ (%)	Ref.
^{238}U sf	0.048(016)	0.159(023)	0.002(002)	0.076(012)	0.595(008)	0.832(007)	8.0(3.7)		Wetherill (1953)
	0.034(003)	0.128(011)	<0.001	0.087(012)	0.605(008)	0.857(007)	6.6(0.6)	6.3	Young & Thode (1960)
			0.002(002)	0.080(008)	0.595(012)	0.835(007)			Shukolyukov & Ashkinadze (1968)
	0.033(004)	0.150(012)	0.002(001)	0.082(002)	0.579(006)	0.825(003)			Shukolyukov (1970)
	0.064(014)	0.218(011)	0.005(003)	0.108(002)	0.580(003)	0.832(004)	5.5(1.7)		Sabu (1971)
	0.046(013)	0.110(020)	0.006(003)	0.088(002)	0.568(003)	0.828(004)	6.0(0.4)		Hebeda (1987)
	0.049+001	0.160+003		0.093+001	0.602+004	0.843+007			
	-010	-015	<0.002	-006	-006	-012			Hünemohr (1989)
	0.042(001)	0.0152(001)	<0.001	0.083(002)	0.570(003)	0.818(003)	6.1(0.1)		Eikenberg et al. (1993)
				0.087(007)	0.578(013)	0.827(008)	6.5(0.5)	**	Ragettli (1993); Ragettli et al., (1994)
^{238}U nf(fast)	0.309	0.634	0.14	0.454	0.73	1.12	4.28	5.9	Maeck et al. (1975)
	0.309	0.637	0.399	0.4877	0.778	1.132	5.1		Crouch (1977)
^{235}U nf(th)	0.27	0.5	0.15	0.453	0.677	1.25	3.17	6.47	Farrer & Tomlinson (1962)
	0.272	0.508	0.0912	0.4448	0.685	1.232	3.2		Crouch (1977)
^{232}Th nf(fast)	0.331	0.608		0.287	0.508	0.952	0.942	5.65	Kennet & Thode (1957)
^{244}Pu sf			<0.026(042)	0.251(022)	0.876(031)	0.921(027)			Alexander et al., (1971)
**	$^{136}\text{Xe}/^{86}\text{Kr} \lambda_{sf} = 6.83 \pm 0.18 \times 10^{-18}$								

where P_{th} is the probability of neutrons being thermalized (0.8), N , the crustal neutron production rate (for upper crust = $11.2 \text{ neutrons g}^{-1} \text{ yr}^{-1}$, Table 3, Eqn. 10), F_{235} , the probability that ^{235}U will capture a thermal neutron that induces fission, and $^{136}\text{Y}_{nf}$, the fraction of the resulting fission product that will form ^{136}Xe (0.0647, Table 7). The thermal neutron-capture cross-section for a fission induced reaction on ^{235}U is 586 barns (Lide 1994). Average upper crust, at 2.8 ppm U with a ^{235}U natural abundance of 0.720%, therefore gives a neutron capture probability of $5.03 \times 10^{-8} \text{ mol barns}$. Total capture probability for average upper crust is 0.00979 mol barns (Table 1) to give $F_{235\text{U}} = 5.13 \times 10^{-6}$, and $^{136}\text{Xe}_{nf} = 2.96 \times 10^{-6} \text{ atoms g}^{-1} \text{ yr}^{-1}$. Therefore the upper crust has $^{136}\text{Xe}_{sf}/^{136}\text{Xe}_{nf} = 1.62 \times 10^4$. Given such a high production ratio, it is indeed surprising to find resolvable $^{136}\text{Xe}_{nf}$ at all. Indeed, there are no reported $^{136}\text{Xe}_{nf}$ components in any accessory phase minerals (e.g., Ragettli et al. 1994; Wieler and Eikenberg 1999). The observation of $^{136}\text{Xe}_{sf}/^{136}\text{Xe}_{nf}$ values approaching unity in massive U-oxides such as uraninite and pitchblende (Fig. 8) (Eikenberg et al. 1993) may reflect one of the few examples of elemental heterogeneity affecting the production rate of thermal neutron induced reactions: Nevertheless, while the density of uranium oxides are larger than the average crust, the elemental neutron stopping power is lower and it is not clear that the neutrons are more efficiently stopped in UO_2 than in average crust pointing therefore, not to mineral scale heterogeneity, but perhaps high concentrations of the minerals on the meter scale. In such a case, many neutrons produced within the U oxide-rich 'layer' or deposit will be thermalized before escaping the U-rich environment, enhancing the rate of production of fission products from ^{235}U compared to average crust. There is no evidence that this is a significant process on a crustal scale.

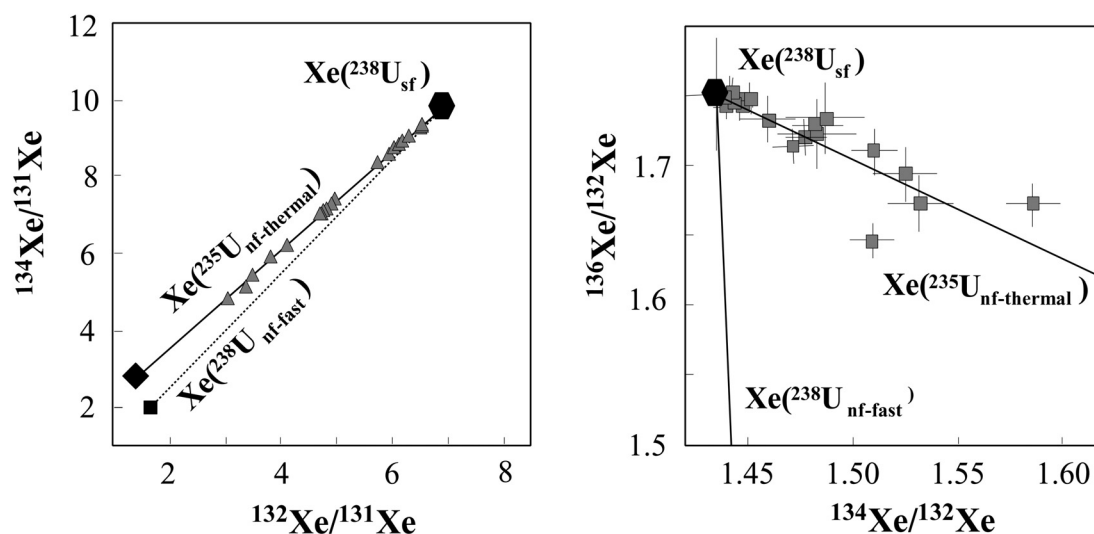


Figure 8. Xenon isotopes corrected for atmosphere contributions measured in uraninite and pitchblendes plotted after Eikenberg et al. (1993). Xe isotopes are dominated by production from the spontaneous fission of ^{238}U ($\text{Xe}(^{238}\text{U}_{sf})$). In these massive samples only, Xe from thermal neutron induced fission of ^{235}U ($\text{Xe}(^{235}\text{U}_{nf-thermal})$) is also resolvable. Xe fission products from fast neutron interactions with ^{238}U ($\text{Xe}(^{238}\text{U}_{nf-fast})$) are negligible.

CFF-Xe. Isotopic Xe anomalies, characterized by enriched ^{129}Xe , ^{131}Xe and ^{132}Xe relative to fission ^{136}Xe , were identified within minerals from the natural nuclear reactor no.2 in the Oklo uranium mine, Gabon (Shukolyukov et al. 1976). Although early theories focused on a nuclear process to account for this observation, it later became apparent that this was a chemical fractionation effect (Meshik and Shukolyukov 1986; Meshik 1988). CFF-Xe, or Chemically Fractionated Fission-Xenon is created when fast

diffusion of radioactive Xe β^- -precursors occurs in fine-grained minerals at high temperature. These fission fragments (Xe, Sn, Sb, Te, I, Kr, Se and Br) migrate from the parent U-bearing mineral into adjacent phases where they subsequently decay to stable Xe isotopes. The longer the lifetime of the Xe precursor (e.g., Sb, Te and I), the larger the isotopic anomaly in the stable Xe daughter-product (Meshik 1988; Meshik et al. 1996).

This process of chemical separation of fission Xe-precursors accounts for Xe and Kr anomalies from the Alamogordo nuclear test site (Pravdivtseva et al. 1986), U-bearing minerals (Gerstenberger 1980; Meshik 1988) and has also been used to explain the isotopic structure of Xe-H in meteoritic nano-diamonds (Ott 1995; 1996). Takaoka (1972) and Igarashi (1995) have compared an assumed fractionated meteorite component (Q or 'planetary') with the atmosphere Xe isotopic composition to resolve the additional atmosphere Xe fission components. The inferred fission spectrum is more enriched in ^{131}Xe and ^{132}Xe than either ^{244}Pu or ^{238}U spectra. Meshik et al. (2000) show that CFF-Xe from Okelobondo (the southern extension of Oklo) matches the theoretical fission composition calculated by Igarashi (1995). While Meshik et al. (2000) argue that the atmosphere Xe isotopic composition can therefore be accounted for with terrestrial CFF-Xe, this appears difficult to accommodate on mass balance grounds: The total fissiogenic ^{136}Xe in the atmosphere is $2.8 \pm 1.3\%$ of the atmosphere ^{136}Xe budget (Igarashi 1995). Only 0.47% of the atmosphere ^{136}Xe can be accounted for by ^{238}U fission in the crust (Table 8). From analogy with ^{40}Ar (see the section: *Production rates today and over the history of the Earth*), only an additional $\sim 20\%$ can be sourced from ^{238}U decay from the mantle to account for, at most, 0.56% of the atmosphere ^{136}Xe budget. Even if this were all CFF-Xe, it could not account for the fissiogenic Xe component in the atmosphere. Furthermore, the total terrestrial fissiogenic Xe produced is unchanged. This scenario would require preferential release of CFF-Xe and efficient retention of the complementary enriched ^{136}Xe component, further degrading the mass balance. There is no reason to believe this occurs. Terrestrial CFF-Xe, while important on a mineral scale, probably has little impact on the global scale. This does not rule out an accretionary CFF-Xe contribution to the Earth's isotopic Xe composition (e.g., Shukolyukov et al. 1994).

Cosmogenic noble gas production

Noble gases are produced by spallation reactions when high energy particles, produced by interactions between cosmic rays and the Earth's atmosphere, interact with nuclei at the Earth's surface (Niedermann 2002, this volume). The purpose of this section is to evaluate how production and preservation of cosmogenic noble gases affect the crustal noble gas budget.

Of the cosmogenic noble gas isotopes that are produced at the Earth's surface, ^3He is produced most rapidly, for example $(^3\text{He}/^{21}\text{Ne})_{\text{cos}} = 4.4$ (Schäfer et al. 1999) and $(^3\text{He}/^{38}\text{Ar})_{\text{cos}} \approx 1.5$ for $[\text{Ca}] = 3\%$ and $[\text{K}] = 2.8\%$ (Renne et al. 2001; Lal 1991). However, the effect of cosmogenic ^3He on the crustal ^3He budget is difficult to assess because ^3He is also lost more rapidly than the heavier noble gases; for example, most common crustal minerals (e.g., quartz, feldspar, clay minerals, carbonates) will not preserve a significant fraction of the ^3He produced. Only olivine, clinopyroxene and certain garnets have been conclusively shown to preserve cosmogenic ^3He over geologically significant periods (Dunai and Roselieb 1996; Kurz et al. 1988; Trull et al. 1991).

The column-integrated ^3He production at the surface at the present day due to reactions induced by cosmic-ray particles is 23×10^3 atoms $^3\text{He} \text{ cm}^{-2} \text{ y}^{-1}$ based on 110 atoms $^3\text{He} \text{ g}^{-1} \text{ y}^{-1}$ at the surface and a mean attenuation length of 160 g cm^{-2} (Niedermann 2002). Given the uncertainties indicated by Niedermann, this value is subject to an error of $\sim 20\%$. The vast majority of this is not preserved because it is produced in minerals that have high He diffusivities at surface conditions; even those minerals that do preserve

Table 8. Current production rate and total noble gas contribution from the crust over the age of the earth.

<i>Present day Production rates cm³ STP/yr</i>										<i>Total Produced in the Crust Over 4.5Ga cm³ STP</i>				
Crust Mass (g)	Oceanic	Lower	Middle	Upper	Total	Cosmo-genic	IDP	Oceanic	Lower	Middle	Upper	Cosmo-genic	Total	% of Atmosp. (O&P'83)
³ He	-	3.30E+03	1.42E+04	4.26E+04	6.00E+04	1.29E+03	7.50E+02	5.00E+24	8.33E+24	9.63E+24	6.13E+24	5.80E+12	4.49E+14	1549%
⁴ He	4.55E+10	4.88E+11	3.54E+12	3.95E+12	7.98E+12	1.29E+03	-	3.58E+20	3.37E+21	2.63E+22	2.93E+22	5.80E+12	5.93E+22	285484%
²⁰ Ne	-	1.96E+03	1.60E+04	1.74E+04	3.53E+04	2.93E+02	1.50E+04	-	1.35E+13	1.18E+14	1.29E+14	1.32E+12	2.62E+14	0.0004%
²¹ Ne	2.28E+03	2.04E+04	1.65E+05	1.79E+05	3.65E+05	2.93E+02	-	1.79E+13	1.41E+14	1.22E+15	1.33E+15	1.32E+12	2.71E+15	1.4051%
²² Ne	-	4.09E+04	3.30E+05	3.58E+05	7.29E+05	2.93E+02	1.20E+03	-	2.82E+14	2.45E+15	2.66E+15	1.32E+12	5.39E+15	0.0810%
³⁶ Ar	-	2.64E+03	9.70E+03	4.49E+04	5.73E+04	7.31E+02	-	-	1.82E+13	7.19E+13	3.33E+14	3.29E+12	4.26E+14	0.0003%
³⁸ Ar	-	1.47E+03	1.19E+04	1.29E+04	2.63E+04	7.31E+02	-	-	1.02E+13	8.81E+13	9.56E+13	3.29E+12	1.97E+14	0.0008%
⁴⁰ Ar	1.17E+10	1.61E+11	6.24E+11	6.72E+11	1.46E+12	7.31E+02	-	2.34E+20	3.23E+21	1.25E+22	1.35E+22	-	2.95E+22	79.94%
⁸³ Kr	9.69E-01	6.87E+00	6.35E+01	7.08E+01	1.41E+02	-	-	6.30E+09	4.47E+10	4.13E+11	4.61E+11	-	9.25E+11	0.0002%
⁸⁴ Kr	3.51E+00	2.49E+01	2.30E+02	2.56E+02	5.11E+02	-	-	2.28E+10	1.62E+11	1.50E+12	1.67E+12	-	3.35E+12	0.0001%
⁸⁶ Kr	2.31E+01	1.64E+02	1.51E+03	1.68E+03	3.36E+03	-	-	1.50E+11	1.06E+12	9.84E+12	1.10E+13	-	2.20E+13	0.0028%
¹³¹ Xe	1.30E+01	9.25E+01	8.55E+02	9.53E+02	1.90E+03	-	-	8.49E+10	6.02E+11	5.57E+12	6.20E+12	-	1.25E+13	0.0170%
¹³² Xe	8.67E+01	6.14E+02	5.68E+03	6.33E+03	1.26E+04	-	-	5.64E+11	4.00E+12	3.70E+13	4.12E+13	-	8.27E+13	0.0892%
¹³⁴ Xe	1.24E+02	8.79E+02	8.13E+03	9.06E+03	1.81E+04	-	-	8.07E+11	5.72E+12	5.29E+13	5.89E+13	-	1.18E+14	0.3297%
¹³⁶ Xe	1.50E+02	1.06E+03	9.83E+03	1.10E+04	2.18E+04	-	-	9.76E+11	6.92E+12	6.40E+13	7.13E+13	-	1.43E+14	0.4690%
neutrons/g/yr		1.05	6.19	11.21	-									

Notes

Surface area and mass of continental crust is 1.5E+8 km² and 2.41E+25 g respectively

Relative mass of crustal reservoirs calculated using Table 9 in Rudnick and Fountain (1995).

Mass of Oceanic crust calculated assuming surface area of 3E+8 km², density=2.8g/cm³ and average depth=6km

Oceanic crust taken to be U = 0.047ppm; Th=0.120ppm; K=600ppm (Sun and McDonough, 1989).

²¹Ne production calculated using Equation 15 and compositions in Table 2. ²¹Ne production will be ~20% higher if observed ⁴He/²¹Ne relationship is used (Fig. 7)²¹Ne/²²Ne crust = 0.5 ³⁸Ar crust = 0.2*²²Ne*Cl/F (Cl/F = 0.18) Cosmogenic ²¹Ne=²⁰Ne=²²Ne Cosmogenic ³He=⁴He IDP ²⁰Ne/²²Ne=12.5

cosmogenic ^3He (olivine, clinopyroxene and garnet) release their cosmogenic ^3He at comparatively low temperatures. Consequently, it is highly unlikely that accumulation of cosmogenic ^3He is a major crustal component.

All three isotopes of Ne are produced at almost equal rates by cosmogenic particles at the Earth's surface. As ^{21}Ne is the least abundant isotope, cosmogenic production of ^{21}Ne is more likely to be a significant portion of the crustal ^{21}Ne budget than for ^{20}Ne or ^{22}Ne . Similarly, ^{38}Ar is the least abundant isotope of Ar as well as the most rapidly produced (for typical crustal compositions), therefore the effects of cosmogenic ^{38}Ar accumulation on the crustal noble gas budget will be examined.

The rate of cosmogenic ^{21}Ne production is comparatively well constrained and the column-integrated ^{21}Ne production at the Earth's surface is $\approx 5 \times 10^3 \text{ atoms cm}^{-2} \text{ y}^{-1}$ ($(^3\text{He}/^{21}\text{Ne})_{\text{COS}} = 4.4$ (Schäfer et al. 1999)). Production of cosmogenic ^{38}Ar at the Earth's surface is less well known: Lal (1991) estimated the rate of ^{38}Ar production from ^{40}Ca to be $\sim 200 \text{ atoms g}^{-1} \text{ Ca y}^{-1}$ and analyses of cosmogenic noble gases in meteorites suggest that ^{38}Ar production from ^{39}K will be about ten times higher than that off ^{40}Ca (Freundel et al. 1986); other sources of cosmogenic ^{38}Ar are unlikely to be important relative to Ca and K production for typical upper crustal compositions. Therefore the $(^3\text{He}/^{38}\text{Ar})_{\text{COS}}$ at the surface ≈ 1.5 although this is uncertain to more than a factor of 2. Typical upper crustal compositions (Ca = 3%; K = 2.8%), will result in a column-integrated ^{38}Ar production rate of $13 \times 10^3 \text{ atoms cm}^{-2} \text{ y}^{-1}$.

Given a land surface area of $150 \times 10^6 \text{ km}^2$, the global annual production of ^{21}Ne and ^{38}Ar is of the order 300 and 700 $\text{cm}^3 \text{ STP}$ respectively (Table 8). Under the assumption of constant cosmogenic production rates and constant land surface area, the total cosmogenic ^{21}Ne and ^{38}Ar production for the Earth is of the order 1 and $3 \times 10^{12} \text{ cc STP}$ respectively, equivalent to $<5\%$ of the nucleogenic production of these isotopes. Given that both the flux of cosmogenic neutrons at the Earth's surface and the land surface area vary with time, the uncertainties on these estimates likely exceed one order of magnitude. However, even given these uncertainties, it is clear that cosmogenic Ne and Ar do not constitute a major contribution to the crustal system.

Interplanetary dust accumulation

While not strictly produced in the crust, it is convenient to consider noble gases from this source here. Oceanic sediments have $^3\text{He}/^4\text{He}$ ratios that are higher than expected for *in situ* radiogenic/nucleogenic production and higher than can be plausibly attributed to detrital mantle – derived material. Accretion of interplanetary dust particles (IDPs) that have high $^3\text{He}/^4\text{He}$ ratios and high He concentrations likely results in high $^3\text{He}/^4\text{He}$ ratios in slow accumulating oceanic sediments. Again, this section is not intended to review the IDP literature, but rather assess the possible impact of IDPs on the composition of the crust as a reservoir for noble gases.

It is known that IDPs can preserve their extra-terrestrial He over long time periods, certainly up to 70 Ma (Farley 1995) and possibly as long as 450 Ma (Patterson et al 1998). As a consequence of their ability to preserve ^3He over long periods of time, accumulated IDPs could represent a significant fraction of the ^3He inventory of the crust. The average IDP flux over the last 70 Ma is equivalent to $0.5 \times 10^{-12} \text{ cm}^3 \text{ STP } ^3\text{He cm}^{-2} \text{ ky}^{-1}$ (Farley 1995). If all the IDPs falling on continents were preserved, this corresponds to a ^3He IDP flux to the continental crust of the order $750 \text{ cm}^3 \text{ STP y}^{-1}$, or about 1% of the ^3He produced in the crust by $\text{Li}(n,\alpha)$ reactions.

Few data exist on the heavy noble gas IDP flux to the Earth. Measurements by Matsuda (1990) suggest the average $^3\text{He}/^{20}\text{Ne}$ of IDPs is in the range 0.05, implying a continental ^{20}Ne flux of $\sim 15,000 \text{ cm}^3 \text{ STP y}^{-1}$, or about 30% of the ^{20}Ne produced in the

crust. IDPs have high $^{20}\text{Ne}/^{21}\text{Ne}$ and $^{20}\text{Ne}/^{22}\text{Ne}$ ratios and therefore are unlikely to impact the crustal budget for ^{21}Ne or ^{22}Ne .

Uncertainties in the IDP flux over the age of the Earth preclude any realistic estimate of the total IDP noble gas inventory. It should also be noted that there is considerable uncertainty in the fate of IDPs falling onto continents. Noble gases, particularly ^3He , are preserved only in IDPs $\leq 50\text{-}\mu\text{m}$ diameter due to frictional heating during entry (Farley 1995). Small particles will be preferentially transported off the continents into oceans, considerably reducing their contributions to the continental noble gas reservoir. The noble gas carrier phase or phases in IDPs is also unknown, and their preservation during crustal processes such as diagenesis and metamorphism is speculative at this time.

Production rates in continental crust today and over the history of the Earth

Having established the factors governing the rates of noble gas production in geological materials, it is a fairly straightforward step to use current estimates of the continental crust chemical composition to calculate the present day radiogenic and nucleogenic production rates in the crust (Table 8). We have used the estimates of the continental crust chemical composition and structure following Rudnick and Fountain (1995) (Table 2). These workers have divided the crust into three portions, lower, middle and upper crust, based on observations from seismic studies summarized by Holbrook et al. (1992). The upper crust (depths from 0 to 10-15 km) is dominated by evolved rock types and contains some 66% of the heat producing elements, U, Th and K, relevant to noble gas production. The middle crust (between depths of 10-15 and 20-25 km) is composed of rocks in the amphibolite facies and is intermediate in composition between the upper and lower crust. The latter is dominated by metamorphic rocks in the granulite facies (between 20- to 25- and 40-km depth) and contains some 6% of the heat producing elements. The decrease in heat producing element concentration with depth is a result of depletion in felsic rocks caused by granulite facies metamorphism and an increase in the proportion of mafic rocks. In total the crust accounts for 42 and 32% of the U and K in the bulk silicate Earth (McDonough and Sun 1995).

Although not explicit in their estimate of the total mass of these crustal subdivisions, the average crust concentration of elements and the individual subdivision concentrations (Table 9 in Rudnick and Fountain 1995) enable us to estimate the relative proportion of these reservoirs. Taking a crustal mass of 2.41×10^{25} g (e.g., Mason and Moore 1980), this gives an upper, middle and lower crustal mass of 8.33×10^{24} g, 9.63×10^{24} g and 6.13×10^{24} g respectively. ^4He and ^{40}Ar production rates are directly proportional to the reservoir (U+Th) and K concentration respectively. ^{136}Xe , ^{86}Kr and related fission product production rates are directly proportional to the U concentration in each reservoir. The products of thermal neutron reactions only, ^3He and ^{36}Ar , are a function of the neutron production rate determined by the U+Th concentration and average chemical composition following Equation (10). The production rate of ^{20}Ne and ^{21}Ne by (α, n) and fast neutron reactions is a function of the U, Th, O, Mg and Na concentration, and we have used the empirical relationship developed in Equations (15) and (16) to estimate their production rates. It should be noted that this value is based on the theoretical production rates, the use of the observed crustal $^4\text{He}/^{21}\text{Ne}$ production ratio (Fig. 7) will result in a $\sim 20\%$ increase in the rate of ^{21}Ne production. We have not used Equation (17) (which is valid for mineral scale compositions) to estimate ^{22}Ne production because of the impact of mineral scale fluorine heterogeneity on the crustal system ^{22}Ne production rate. We have assumed for all crustal reservoirs a $^{21}\text{Ne}/^{22}\text{Ne}$ production rate of 0.52, and based ^{22}Ne production estimates on the ^{21}Ne production rate. Similarly, ^{38}Ar production is dominated by (α, n) reactions on ^{35}Cl , and is likely to be affected by mineral scale heterogeneity. We therefore neglect Equation (25) (which is valid for mineral scale

compositions). We have assumed that the average crustal Cl/F ratio (0.18) is preserved on the mineral scale to give a ^{38}Ar production linked to the ^{22}Ne rate where: $^{38}\text{Ar} = 0.2 \times 0.18 \times ^{22}\text{Ne}$.

To place some of these production rates into perspective we can consider the air-derived concentration of selected noble gases dissolved in groundwater. 1 g of water contains approximately $4.6 \times 10^{-6} \text{ cm}^3 \text{ STP } ^4\text{He}$, $6 \times 10^{-10} \text{ cm}^3 \text{ STP } ^{21}\text{Ne}$, $2 \times 10^{-8} \text{ cm}^3 \text{ STP } ^{22}\text{Ne}$, $2.5 \times 10^{-7} \text{ cm}^3 \text{ STP } ^{38}\text{Ar}$, $3.9 \times 10^{-4} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$, $1.6 \times 10^{-8} \text{ cm}^3 \text{ STP } ^{86}\text{Kr}$, $1.2 \times 10^{-9} \text{ cm}^3 \text{ STP } ^{136}\text{Xe}$. 1 g of rock at 1% porosity can contain $\sim 3.5 \times 10^{-3} \text{ g}$ of groundwater. Based on present day upper crust production rates a 10% crustal contribution to these isotopes will occur in $3.9 \times 10^3 \text{ yr}$ (^4He), 7.2 Ma (^{21}Ne), 120 Ma (^{22}Ne), 42 Ga (^{38}Ar), 1.2 Ma (^{40}Ar), 20 Ga (^{86}Kr) and 234 Ma (^{136}Xe). Clearly ^4He is the most sensitive indicator of crustal fluid addition to near surface reservoirs, and can be used as a groundwater dating tool (Kipfer et al. 2002, this volume). Over longer, but similar, timescales ^{21}Ne and ^{40}Ar contributions will be observed, while systems that show significant crustal ^{22}Ne contributions should also contain resolvable fissiogenic ^{136}Xe . Only crustal systems with no groundwater or air contamination can hope to show ^{38}Ar or ^{86}Kr variations caused by nuclear processes in the crust.

Although the rate of crustal growth remains controversial (e.g., Armstrong 1991; Taylor and McLennan 1995), the total amount of radiogenic noble gases produced by the radioelements now in the crust provide an important cornerstone in understanding both the extent to which the silicate Earth has degassed, and the radiogenic-nucleogenic noble gas contribution to the noble gas budget of the atmosphere. While, the compatibility of noble gases relative to the heat producing elements is uncertain in some melting regimes (e.g., Chamorro-Perez et al. 1998; Barfod et al. 2002), both are highly incompatible during melting at mid-ocean ridges (e.g., Marty and Lussiez 1993), and noble gases are insoluble in silicate melts (Carroll and Draper 1994). The end result will be that noble gases produced within the mantle by radioelements or their daughter products which are now incorporated in the continental crust, will have been degassed into the atmosphere (e.g., Coltice et al. 2000). The efficiency of release of noble gases produced in the crust is discussed in some detail later in this contribution, but the concept of 'average crustal K- ^{40}Ar age' provides an indication of total crust degassing efficiency. This is estimated to be $1 \times 10^9 \text{ yr}$ (Hurley and Rand 1969). The K in the continental crust over 4.5 Ga will have produced some $2.92 \times 10^{22} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ (Table 8). In the last $1 \times 10^9 \text{ yr}$ the crust will have produced $1.95 \times 10^{21} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ or 6.7% of the total continental crustal production. The 'average crustal K- ^{40}Ar age' of $1 \times 10^9 \text{ yr}$ therefore suggests that the crust retains only 6.7% of its ^{40}Ar .

Of the $7.60 \times 10^{22} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ produced by K decay in the bulk silicate Earth, $2.95 \times 10^{22} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ has been produced in the continental and oceanic crust (Table 8). Deducting $1.95 \times 10^{21} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ for ^{40}Ar still trapped, crustal production accounts for 75% of the $3.69 \times 10^{22} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ now in the atmosphere. $1.24 \times 10^{20} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ in the atmosphere is from primordial sources, assuming primordial $^{40}\text{Ar}/^{36}\text{Ar} = 1$ (Ozima and Podosek 1983). Some $9.23 \times 10^{21} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ in the atmosphere must therefore be derived from decay of K still in the mantle, or crustal K that has been recycled (e.g., Coltice et al. 2000). $3.72 \times 10^{22} \text{ cm}^3 \text{ STP } ^{40}\text{Ar}$ must still reside in the mantle. Taking the mass of the mantle to be $4.02 \times 10^{27} \text{ g}$, this would require an average mantle concentration of $9.27 \times 10^{-6} \text{ cm}^3 \text{ STP } ^{40}\text{Ar/g}$, some 4.2 times higher than the value of $\sim 2.2 \times 10^{-6} \text{ cm}^3 \text{ STP } ^{40}\text{Ar/g}$ estimated from MORB-source ^4He concentrations and a $^4\text{He}/^{40}\text{Ar}^*$ ratio of 2 (where $^{40}\text{Ar}^*$ is the non-atmospheric ^{40}Ar contribution assuming that all ^{36}Ar is atmospheric in origin; see Porcelli and Ballentine 2002). This forms one of the key arguments for a region of the mantle that is less degassed than the mid ocean ridge source (e.g., Allègre et al. 1996). Calculations that investigate radiogenic contributions to

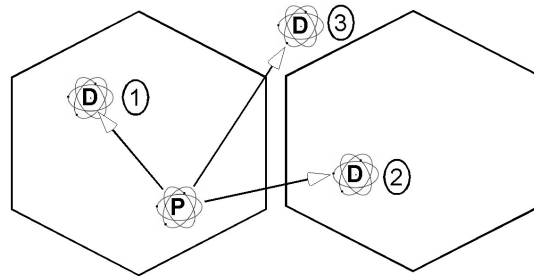
the Earth's atmosphere must take the mantle degassing efficiency into consideration. For example while the crust contributes $\sim 0.47\%$ of the ^{136}Xe in the atmosphere, assuming the same degassing efficiency for Xe as Ar, the contribution from U still within the silicate mantle can only be 0.11% of the atmosphere ^{136}Xe budget.

RELEASE OF NOBLE GASES FROM MINERALS IN THE CRUST

The $8.0 \times 10^{12} \text{ cm}^3 \text{ STP/yr}$ of ^4He produced in the crust (Table 8) compares with the ^4He flux from the mantle at mid ocean ridges estimated to be $2.2 \times 10^{12} \text{ cm}^3 \text{ STP } ^4\text{He/yr}$ (e.g., Porcelli and Ballentine 2002, this volume). The top 10–15 km of the crust produces some 66% of the crustal radiogenic-nucleogenic budget, 28% in the middle crust (>10 –15 but <20 – to 25-km depth), with only 6% of the production being accounted for from deeper regions of the crust (>20 – to 25-km depth) (Rudnick and Fountain 1995; Tables 2 and 8). Melting and volatile loss at mid ocean ridges efficiently degasses the material that reach this region. The crustal degassing process is very different and controlled by two stages: (1) release from the mineral in which the noble gas was produced/trapped; and (2) transport from the site of production to the surface. It is important to distinguish the relative rates of the underlying processes as very different mechanisms control mineral- and crustal-scale transport of noble gases. Noble gases produced in crustal minerals by U, Th and K decay, or by nuclear reactions with α -particles or neutrons can be released by three main mechanisms: recoil, diffusion or chemical transformation (Fig. 9). We discuss these in turn.

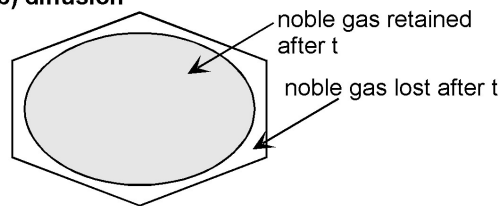
Recoil loss

He and α -recoil. The high kinetic energies of alpha particles produced by U-series decay results in ejection of the alpha particles some distance from their parent nucleus. The distance traveled by the alpha particle—the stopping distance—is determined primarily by the density of the host mineral, although chemistry can be significant. Typical stopping distances are given in Figure 10 from which it is clear that α -particle stopping distances are on the same scale as grain sizes in fine-grained crustal rocks. It is important at this point to distinguish between grain size and effective grain size. This occurs when a single mineral consists of domains, or smaller regions, separated by fast diffusion pathways through the lattice, such as defects in the crystal lattice (see reviews by Farley 2002 and Kelley 2002, both this volume). When the shortest dimension of the grain hosting the parent nuclide approaches the ejection distance, a significant proportion of the He produced will be lost from that grain. Although the He atom may come to rest within any void space between grains, the stopping distance is determined by the density of the medium. Any space between grains will be occupied by low-density fluids. There is a high probability therefore that the He will more likely be buried in the adjacent grain (Rama and Moore 1984). The He atom in this case will come to rest at the end of a damage track that has one end that is “open” to porosity or fast diffusion pathways (as opposed to damage tracks that start and finish within a single grain). He diffusivity is higher along damage tracks (Hurley 1950). He will readily escape from these minerals provided the tracks do not anneal before the He atom can diffuse out of the grain. Annealing temperatures are generally slightly higher than the closure temperature for He diffusivity (e.g., apatite annealing occurs at about 110°C (Laslett et al 1987) whereas He closure in apatite occurs at around 75°C for typical grain sizes and geothermal gradients (Wolf 1996; Farley 2002)). However, fission track annealing data only exist for U-rich phases (mostly titanite, zircon and apatite) and the annealing behavior of most other mineral groups is poorly constrained. If the track does anneal, then the mineral diffusivity controls He escape, otherwise, He is rapidly released from the grains into the void space via the α -damage tracks. Water-filled porosity will be more efficient at stopping α

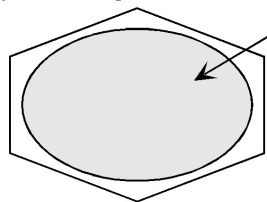
a) recoil

Parent, P, decays to daughter, D, with a characteristic recoil distance in the mineral, X_r . Depending on the grainsize and location of the parent, the recoiled daughter will either come to rest in the same grain (1), or in an adjacent grain (2),

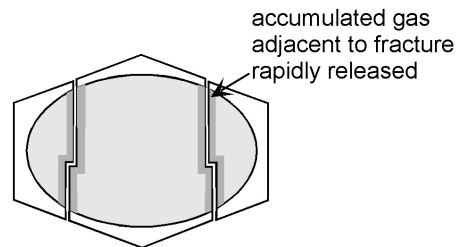
or in the porosity surrounding the grains (3). Note that 1 and 2 are not equivalent as the damage track for 2 has a termination in the porosity. Also, 2 is more probable than 3 because the stopping distance in the fluid between grains will be much greater than that within grains.

b) diffusion

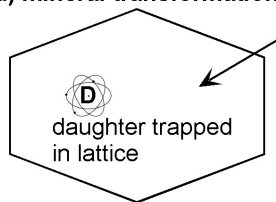
timescale for release depends on diffusivity, grain size and morphology

c) fracturing

before fracturing



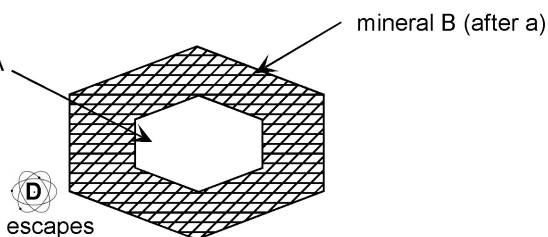
after fracturing: release timescale reduced; accumulated He released rapidly

d) mineral transformation

mineral A

daughter trapped in lattice

daughter escapes (incompatible)



mineral B (after a)

Figure 9. Mechanisms of noble gas release from crustal minerals.

particles than the equivalent dry rock, increasing the fraction of particles that are retained in the pore-space.

The fraction of ^4He lost by recoil is a strong function of effective grain size, or more strictly its the surface/volume ratio (Farley et al. 1996; Martel et al. 1990). Low surface area/volume ratios of $<0.03 \mu\text{m}^{-1}$ result in $>90\%$ retention of ^4He in all but the most

planar or acicular crystal morphologies. As a consequence, recoil loss of ^4He can be expected to decrease with increasing metamorphic grade and crustal depth as average grain sizes generally increase with depth. In contrast, in lithologies showing chemical breakdown and significant re-distribution of U on grain boundaries, recoil loss will form the dominant mechanism of ^4He input into the surrounding fluids (Torgersen 1980; Torgersen and Clarke 1985; Andrews et al. 1989a).

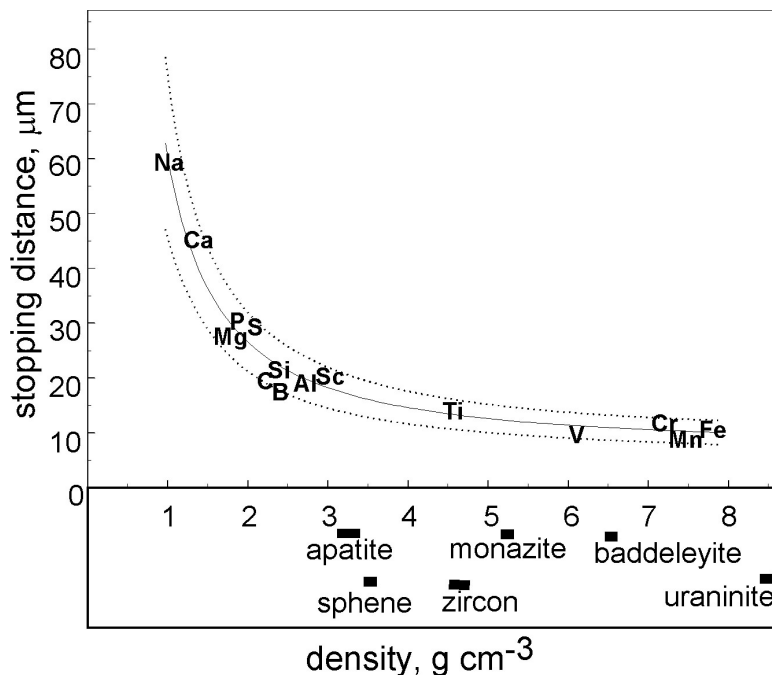


Figure 10. α -particle recoil distances as a function of density for commonly occurring U-rich minerals. The total path length for 5 MeV α -particles has been computed for pure solid elements (up to Fe) following Ziegler (1977); these have been plotted against the density of the pure element above. The stopping power of these elements is primarily a function of their density; therefore the expected stopping power of various minerals can be estimated from the mineral density. The density of common U-bearing phases is shown for comparison. A fitted 2nd order polynomial through the element data (solid line) and 90% confidence interval (dotted lines) are shown. Note that this figure should be used as an approximate guide to α recoil distances only. Significant deviations from these predicted α stopping distances will result from variations in the energies of the emitted α particles as well as deviations from the empirical relationship between density and stopping power. Quantitative stopping distances for specific minerals can be computed according to Ziegler (1985) or using the MDRANGE simulation (Nordlund 1995; see also <http://www.helsinki.fi/~knordlun/range.html>).

Wind-blown sediments. The fine grain size of wind blown sediments ($\leq 8 \mu\text{m}$) precludes significant ^4He production within the sediments themselves as ^4He will recoil out of grains of this size; once a rock has been broken down to a grain size that can be efficiently transported by wind ($\leq 10 \mu\text{m}$), the grains are smaller than the ^4He recoil distance (see Fig. 9). Patterson et al. (1999) demonstrated that cyclic variations in the ^4He concentration (5 to $15 \times 10^{-9} \text{ cc STP g}^{-1}$) of oceanic sediments may be used to trace the wind-blown continental contribution to the deep ocean (loess). Hence, the ^4He in oceanic wind-blown sediments must be due to accumulation of radiogenic He in the continental regions from which the dust was derived. The radiogenic ^4He in the sediments is likely to be located in zircon grains; zircons are known to be He retentive (Farley 2002, this volume), and the remaining minerals (quartz, phyllosilicates) are either U-poor or not He-retentive (or both). Based on measurements of Zr concentrations in

their samples, Patterson et al. estimated that ^4He concentrations in zircons were $\sim 8 \times 10^{-6}$ cc STP g^{-1} , well within the range observed in zircons in general. While having potential as a tracer, volumetrically this source is very small.

Xe and Rn. Minor loss of fissiogenic ^{136}Xe , with a recoil distance $\approx 5 \mu\text{m}$ (Ziegler 1980; Ragettli et al. 1994) can result in Xe loss in fine-grained U-bearing minerals. Recoil of ^{222}Rn from mineral surfaces into groundwaters can be used to estimate groundwater flow rates; while the recoil distance of ^{222}Rn is small ($0.036 \mu\text{m}$), the short half-life of ^{222}Rn (3.8 days) relative to diffusive timescales ensures that the only realistic source of ^{222}Rn is that which is recoiled/immediately released from the minerals in which it was produced. The energy released during production of the remaining radiogenic or nucleogenic noble gas isotopes is sufficiently low to ensure that recoil is small in relation to typical effective grain sizes (Krishnaswami and Seidemann 1988), consequently these nuclides will only be released by diffusion, fracturing or chemical alteration.

Diffusive loss from minerals

Diffusion of noble gases out of minerals plays a direct role in the loss of gas at a mineral scale and is often the underlying process of other release mechanisms such as fracturing or weathering. There are relatively good estimates of the diffusivities of noble gases through different minerals, particularly for systems that have applications to geochronology. A full discussion of diffusion mechanisms and the controls on diffusion can be found in Farley (2002), Kelley (2002) and McDougall and Harrison (1988).

The efficiency of noble gas release from minerals is dependent on D/a^2 (D = diffusivity, a = radius of effective diffusion domain assuming spherical geometry). Increasing the diffusivity for example, by raising the temperature, or decreasing the effective diffusion domain size will increase the rate of noble gas loss through diffusion. Typical rates of He and Ar diffusive loss in crustal minerals are given in Figure 11 and Table 9.

The fraction of gas retained by the mineral phase can be approximated by:

$$F_R = 6/\pi^2 \exp[-\pi^2 Dt/a^2] \quad (28)$$

(after Mussett 1969), where F_R is the fraction of gas retained, D , t and a are diffusivity, time and effective grain size respectively (spherical approximation). At upper crustal conditions ($<150^\circ\text{C}$) He diffusivities in fine-grained (0.1 mm) crustal minerals can range between $10^{-18} \text{ cm}^2 \text{ s}^{-1}$ to $10^{-22} \text{ cm}^2 \text{ s}^{-1}$ (Lippolt and Weigel 1988; Trull et al. 1991). At $10^{-18} \text{ cm}^2 \text{ s}^{-1}$ He is lost from the minerals more rapidly than it can accumulate. Within 10^5 years the rate of production of ^4He equals the rate of loss and the mineral is in 'steady-state' with respect to ^4He production and loss. However, for Ar with diffusivities $<10^{-25} \text{ cm}^2 \text{ s}^{-1}$ (McDougall and Harrison 1988), virtually all Ar is retained in the grains (see Fig. 11). Diffusive release of noble gases from minerals at temperatures where He, but not Ar, is effectively released from the grains will result in high He/Ar ratios in any associated fluid relative to that produced in the mineral (Elliot et al. 1993; O'Nions and Ballentine 1993; Ballentine et al. 1994) (Fig. 12).

The mineral-fluid system and diffusive gradient. For diffusion to occur there must also be a chemical gradient. For noble gases in a mineral-fluid system this occurs when the noble gas equilibrium concentration in the mineral phase (C_m) is higher than the equilibrium concentration in the fluid phase (C_f), where the mineral-fluid partition coefficient (K_d) is defined as

$$K_d = C_m/C_f \quad (29)$$

Values of K_d for He and Ar are not well constrained, although are likely to be $\leq 10^{-3}$ (Brooker et al. 1998; Kelley 2002). Ignoring equilibrium partition coefficients and only

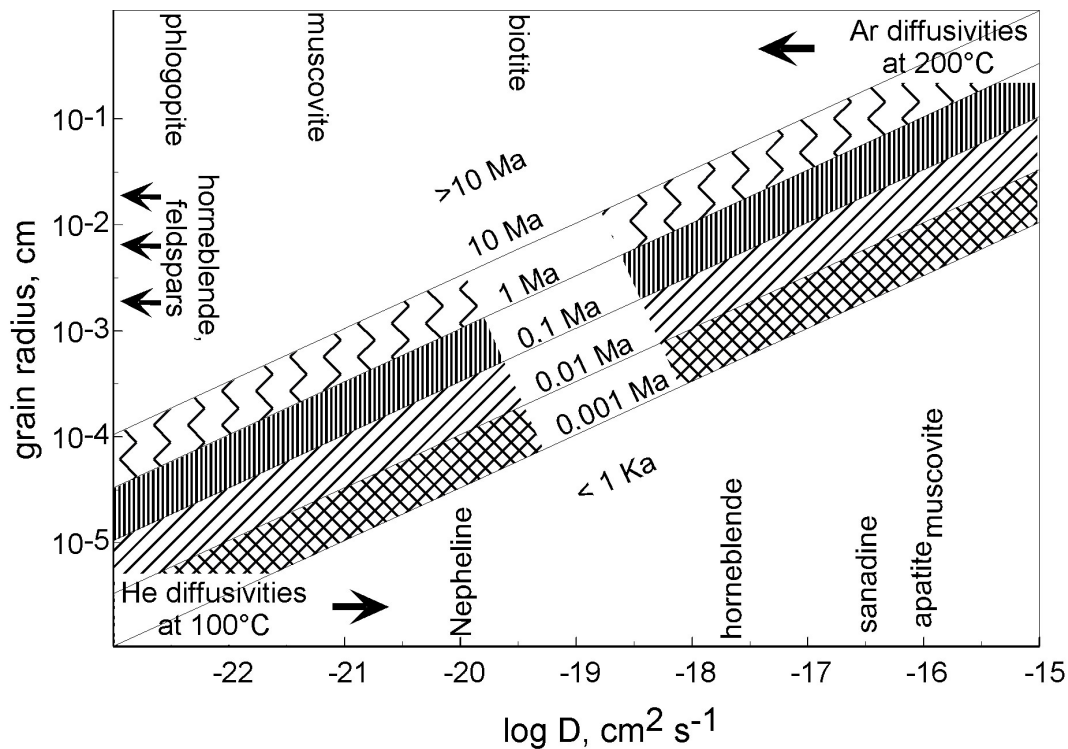


Figure 11. Mineral He and Ar retention is shown as a function of grain size and accumulation time. The time required for a grain to lose 90% of its He or Ar is contoured as a function of mineral diffusivity and grain size, using Equation (30) assuming spherical grains after Mussett (1969); the diffusivities of He at 100°C and for Ar at 300°C are shown for a few representative minerals (data from Lippolt and Weigel 1988; McDougal and Harrison 1988; Wolf et al. 1996). For example, He in a muscovite grain with an effective radius of 100 μm (10^{-2} cm) at 100°C will lose 90% of its He in a few hundred thousand years, while Ar in the same grain is still effectively retained at 300°C over >10 Ma.

considering relative mineral-water concentrations, Andrews et al (1989b) erroneously calculated that typical crustal minerals (~5 wt % K) require in excess of 650 Ma to generate sufficient radiogenic ^{40}Ar to set up a positive diffusion gradient with air-saturated water ($[\text{Ar}] = 2 \text{ to } 5 \times 10^{-4} \text{ cm}^3 \text{ STP } ^{40}\text{Ar g}^{-1}$; Ozima and Podosek 1983). However, when partition coefficients are considered a diffusive gradient between a mineral with 5% K and air saturated water is established within 0.65 Ma (assuming $K_d = 10^{-3}$). He concentrations in air saturated water are much lower than Ar, causing a chemical gradient to be established very rapidly.

In addition to K_d , the diffusion gradient will also be dependent on the mass ratio of solid to fluid, which together govern fluid's ability to provide a suitable sink for the noble gas released. Baxter et al (2001) for example, show that for diffusion gradients to be established:

$$K_d (M_R/M_F) \ll 1 \quad (30)$$

where M_R and M_F are the masses of rock and fluid within the characteristic distance of the production site. Because this characteristic distance is relatively small, $M_R/M_F \approx 1/\text{porosity}$ and therefore if K_d is smaller than the porosity then there will be a diffusion gradient from mineral to fluid. In environments where the porosity is low and the equilibrium concentration in the available porespace is reached, the system will then be controlled by the rate of bulk diffusive loss through the rock porespace. The observation that 'parentless' or 'excess' ^{40}Ar is found in some low porosity systems demonstrates

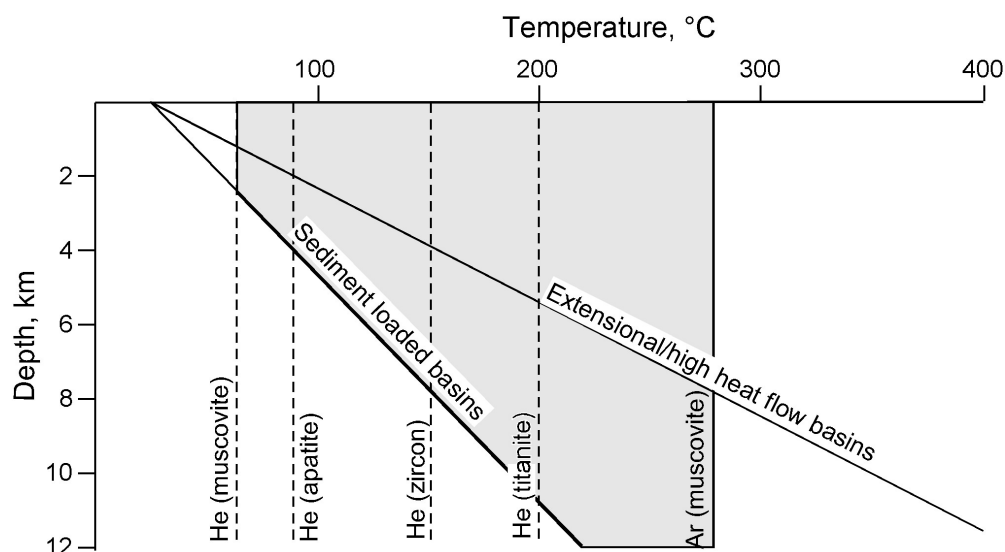


Figure 12. ^4He closure temperatures of common U-rich minerals compared with closure of Ar in muscovite (which has the lowest closure temperature of common K-bearing crustal phases) provides an end-member for considering relative ^4He vs ^{40}Ar release in different thermal regimes (O’Nions and Ballentine 1993; Ballentine et al. 1994). For typical crustal grain sizes and mineralogies, the He closure temperature is lower than that of Ar by over 100°C . If crustal minerals are held at a temperature between the He and Ar closure temperatures then the majority of the He in the rock will be released compared with only a small fraction of the Ar, resulting in elevated He/Ar ratios. Below the He closure temperature, no noble gases are released while above the Ar closure temperature, the $^4\text{He}/^{40}\text{Ar}^*$ ratio will be close to production. This is illustrated above where the likely range of He and Ar closure temperatures are given by the vertical dotted lines while the solid lines show the geothermal gradients of typical loading basins (e.g., Po basin, Italy) and extensional basins (e.g., the Pannonian basin, Hungary). As expected, $^4\text{He}/^{40}\text{Ar}^*$ in the Po basin (up to 124, average = 33) is considerably higher than the production ratio (≈ 5) while $^4\text{He}/^{40}\text{Ar}^*$ of the Pannonian Basin fluids (up to 25, average = 7.7) approach the production value. Similar patterns are found elsewhere: see text for references.

that there are situations where Ar diffusional loss through the rock is slower than Ar production in the mineral. For example, Kelley and Wartho (2000) found ^{40}Ar in kimberlitic phlogopites that required Ar preservation in the phlogopite grains at $\sim 700^\circ\text{C}$. This is 350°C above the phlogopite closure temperature. The phlogopite itself cannot retain Ar at these temperatures in an open system, therefore Kelley and Wartho (2000) concluded that the surrounding rock system must have retained a significant proportion of the ^{40}Ar produced at high temperatures. In this situation, diffusion out of the mineral grains does not occur due to the lack of a diffusion gradient. Instead, the ^{40}Ar will be distributed through the different minerals in the rock according to their respective K_d s, resulting in the ‘parentless’ ^{40}Ar observed in the phlogopite. While there have not been any equivalent studies on the distribution of He, the fact that K_d values are of the same magnitude for He as for Ar (Kelley 2002) suggests that ‘parentless’ He will also be trapped in minerals above the He closure temperature in low porosity environments due to the lack of a diffusion gradient. While these observations have implications for thermochronology, it is likely that the fraction of He or Ar retained in the crust, particularly the upper crust, above their respective closure temperatures is low, and will not represent a significant crustal reservoir for the noble gases.

Fracturing. Dilatant fracturing—the inelastic failure of brittle rocks that occurs during compressional loading—results from microfracturing prior to macroscopic fracturing. By compressing granites and basalts under vacuum, Honda et al (1982) showed that dilatant fracturing of rocks releases noble gases from the minerals in which

Table 9. Frequency factors and activation energies for He and Ar diffusion in selected crustal minerals.

<i>Helium</i>							
<i>Mineral</i>	<i>ref</i>	<i>note</i>	<i>Ea</i> <i>KJ mol⁻¹</i>	<i>err</i>	<i>log(D₀)</i> <i>cm² s⁻¹</i>	<i>err</i>	<i>D (20 °C)</i> <i>cm² s⁻¹</i>
muscovite	1		87	10	-3.94	0.71	3.34E-20
nepheline	1		88	6	-3.94	0.46	2.21E-20
sanidine	1	average	94	4	-3.34	0.4	7.47E-21
hornblende	1	average	104	2	-3.02	0.15	2.56E-22
Langbeinite	1		126	5	-0.12	0.35	2.39E-23
augite avg	1	average	120	8	-1.34	0.5	1.7E-23
hornblende	1		120	4	-1.91	0.34	4.57E-24
nepheline	1		131	5	-2.44	0.4	1.46E-26
apatite	2		152.04	4	5.2	0.5	1.12E-22
granite	3		69.1	4.3	-3.19778		2.9E-16
sediment	4	20°C only					
<i>Argon</i>							
phlogopite	5		243.2	10.9	-0.12494		2.8E-44
biotite		6	197.4		-1.11351		4.31E-37
hornblende		6	269.22		-1.61979		2.01E-50
muscovite		6	168		-6.22185		5.97E-37

References:

- 1: Lippolt and Weigel (1988); 2: Wolf et al (1996); 3: Hussain (1996);
 4: Ohsumi and Horibe (1984); 5: Giletti and Tullis (1974); 6: McDougall and Harrison (1988)

they were produced. The fraction of noble gas liberated from the rocks during compression increased linearly with the dilatant volume change (dilatancy). More He than Ar was lost at any given dilatancy, suggesting that diffusion into and along microfractures created by compression was the dominant release mechanism.

The effect of rock fracturing was examined numerically by Torgersen and O'Donnell (1991), who considered a one-dimensional slab of infinite length (Fig. 13). The gas flux out of the rock increases as a function of the fracture spacing of the slabs before ($= L$) and after ($= l$) a fracture event (for a given diffusivity). The increase in gas flux reaches a maximum of 10^4 times greater flux than the steady-state flux for $L/l = 100$. The timescale of release depends on D/l^2 but is likely to be short (Ka rather than Ma) for geologically reasonable conditions. For example, a 10m fracture spacing will result in loss of all the He that had accumulated prior to fracturing in ~ 1500 yr ($D = 10^{-7} \text{ cm}^2 \text{ s}^{-1}$). However, the dependence of the gas release on D means that He will be released much more rapidly than Ar. Thus, the first fluids released from the slab will have high He/Ar ratios whereas later fluids lost from the slab will have low He/Ar ratios. Fracture release of noble gases can fractionate He from Ar and will likely result in episodic gas release related to tectonic movement in the brittle upper sections of the crust (Torgersen et al 1992b).

Mineral breakdown/diagenesis/metamorphism/alteration

Mineral transformations during diagenesis (e.g., illite $\Rightarrow$ smectite transition), metamorphism (e.g., recrystallization of clay minerals to biotites, amphiboles, etc) or alteration (e.g., serpentinization of mafic minerals) are likely to release the radiogenic noble gases that were produced within their lattices. This assumes that:

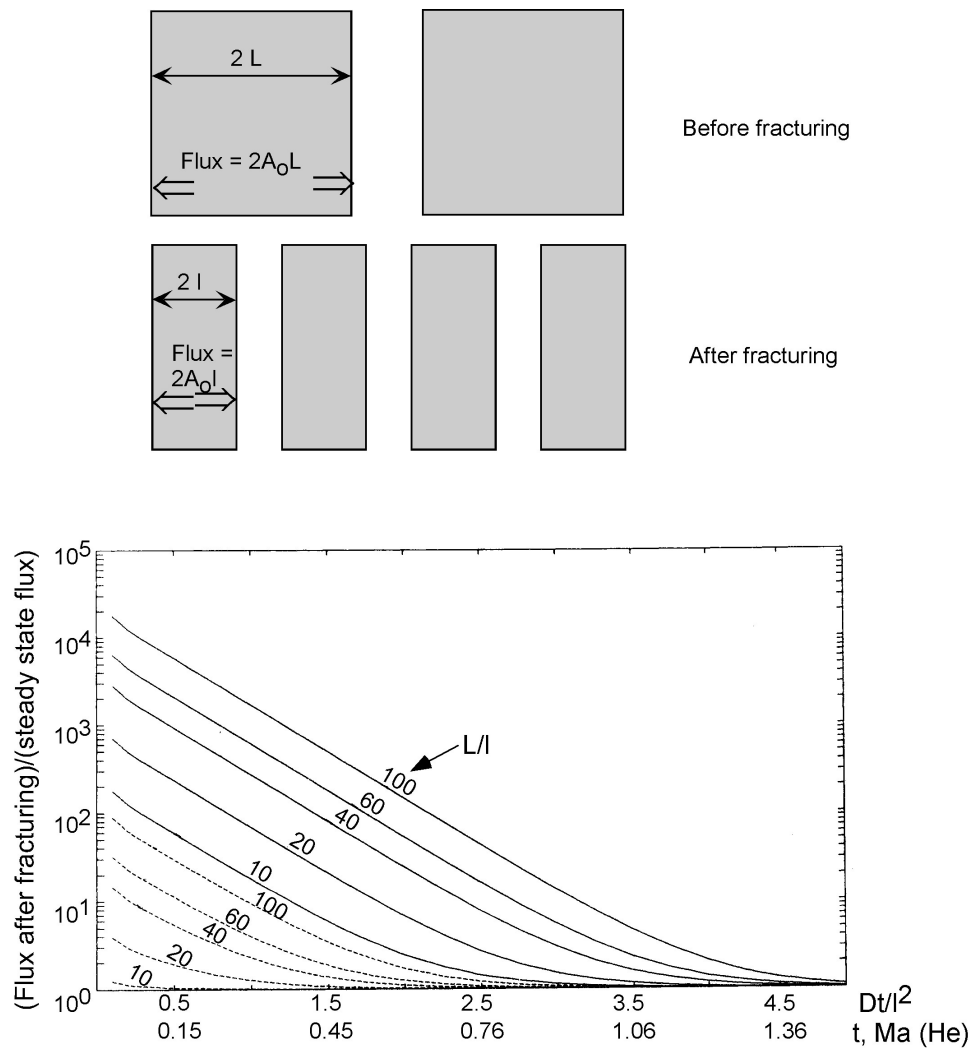


Figure 13. Predicted release of He and Ar from recently fractured rocks (after Torgersen and O' Donnell 1991) TOP: Schematic illustration of the concept behind the numerical model of Torgersen and O'Donnell. Infinite length slabs of half-width L are fractured to a new infinitely long slab of halfwidth l ; the medium between slabs is assumed to be an infinite noble gas sink, therefore the steady-state flux out of the slab is a function only of the noble gas production rate (A_0) and slab thickness. BOTTOM: Computed increases in He or Ar flux (relative to the expected steady-state flux) as a function of time after fracturing. Flux computed for different unfractured/fractured ratios (L/l) for steady-state (solid lines) and open system (dashed lines) initial conditions. Time is given as the non-dimensional parameter Dt/l^2 and as likely He release times: these are calculated for $l = 10$ m and $D_{\text{He}} = 1 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1} \text{ m}^2 \text{ a}^{-1}$. The flux of Ar from the fractured slab follows the same curves as those for He except that the timescale for release scales with $D_{\text{He}}/D_{\text{Ar}}$, which is likely of the order $10^4 - 10^5$, i.e., $Dt/l^2 = 1$ corresponds to 0.3 Ma for He but 3000 Ma for Ar ($D_{\text{Ar}} \approx 1 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$). Therefore, fracturing is likely an efficient mechanism for releasing He from the crust and for producing fluids with high He/Ar, but is unlikely to affect the outgassing of crustal Ar, unless fracturing occurred at relatively high temperatures (and therefore high D_{Ar}).

- (a) noble gases are incompatible elements during these processes; and
- (b) a mechanism exists to remove noble gases from these sites.

While there is robust evidence for the incompatible behavior of all the noble gases (Brooker et al. 1998), there may not always be a suitable pathway for the noble gases to

escape the site of recrystallization. This potentially leads to parentless or 'excess' Ar trapped in minerals (Baxter et al. 2001; Kelley and Wartho 2000; Kelley 2002, this volume). While this is clearly an important problem for Ar-Ar dating on a mineral scale, it seems unlikely in the face of observations of large ^{40}Ar fluxes, from for example the Great Artesian Basin, Australia (Torgersen et al. 1989), that this mechanism can affect large scale degassing of noble gases from the crust.

Unlike diffusive or recoil processes, gas release by mineral transformation is unlikely to elementally fractionate the noble gases, although it is conceivable that the minerals may not have preserved the noble gases at their production ratio. Mineral breakdown at a constant rate will effectively reduce the average mineral grain size and make them more amenable to loss of noble gases via diffusion as discussed above. However, if there is a hiatus in the mineral breakdown, then radiogenic noble gases will accumulate during the period where the minerals are stable. This is unlikely to be a major source of ^4He as He is less likely to accumulate in crustal minerals than Ar owing to the higher rate of He diffusion. Nevertheless, it is possible that, if the accumulation period is significant, noble gas release from mineral breakdown can form the dominant source of radiogenic noble gases in pore fluids.

Whole rock He concentrations from Stripa show that, at depth, the granite retains $\leq 10\%$ of the He produced. The He concentration profile in the surrounding groundwater suggests that diffusive loss of He to the surface via the aqueous phase is the rate limiting step, and is explained if almost all of the ^4He produced is released into the surrounding fluid (Fig. 14; Andrews et al. 1989a). In this system it would appear that He diffusion out of the mineral phase is not the limiting step. Nor can it be explained by ^4He steady state release from minerals. High ^{222}Rn concentrations in these fluids require that some of the Rn produced within the granite is released from its' production site into the fluid on timescales similar to $t_{1/2}^{222}\text{Rn} = 3.5$ days (Andrews et al. 1989a), and He release timescales are likely to be similar to those of Rn. Andrews et al. argue that the noble gases found in the Stripa groundwaters can be accounted for by the chemical breakdown of the minerals in which they were released. This explains the rapid release of both He and Rn into the surrounding fluid. Further evidence is provided by resolvable radiogenic Ar concentrations in the Stripa fluids (Andrews et al. 1989a), inconsistent with diffusion as the dominant mechanism for releasing noble gases from crustal minerals (e.g., Elliot et al. 1993). Similar conclusions have been made for the Carnmenellis granite, England (Andrews et al. 1989a; Martel et al. 1990).

Additional arguments for local metamorphic reactions as a significant mechanism for releasing noble gases from minerals comes from Ar-Ar dating studies that show there are lithological controls on 'parentless' ^{40}Ar ; for example, 'parentless' ^{40}Ar is correlated with K content in some charnockites (Foland 1979) and eclogites (Scaillet 1996). Because the equilibrium mineral-fluid partition coefficient for noble gases, K_d , is very low, if a fluid is present it is unlikely that the released noble gases will be re-incorporated in the newly formed minerals. However, in the absence of a significant fluid volume (Eqn. 30) (e.g., in the lower crust), the released gases must re-distribute themselves depending on their solubilities in the respective new mineral phases (Baxter et al. 2001).

TRANSPORT OF NOBLE GASES FROM THE DEEP CRUST TO SHALLOW LEVEL SYSTEMS

Release of noble gases from minerals by recoil, diffusion, fracturing or chemical change defines the amount of crustal radiogenic-nucleogenic noble gas and their relative abundance released into the proximal fluid system. The transport of the noble gases from

the proximity of their production to near surface systems and, ultimately, the atmosphere, is dependant on the driving force. This must be in the form of either a concentration gradient driving diffusion, or a pressure gradient resulting in advective fluid flow. With geologically reasonable timescales and parent element concentrations, noble

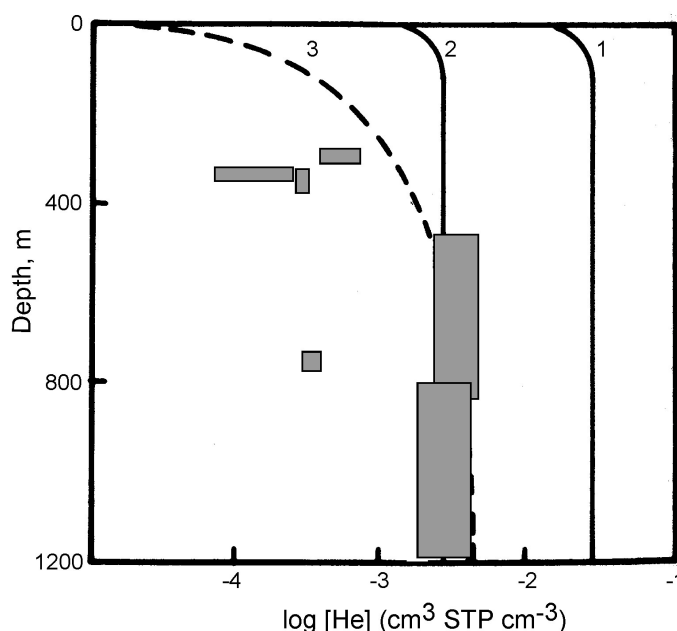


Figure 14. He concentration profiles through the shallow crust (after Andrews et al. 1989a). The He concentration profiles expected in groundwaters for He diffusion through bulk granite (solid lines) with diffusion coefficients of $1.6 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$ ($5 \times 10^{-7} \text{ m}^2 \text{ a}^{-1}$) assuming that either all the He produced is available for diffusion (profile 1) or that 10% of the He produced is available for diffusion (profile 2). The dashed line shows the profile expected if all the He produced diffuses from the granite into the groundwater, followed by diffusive loss from the groundwater to the atmosphere. The boxes show the compositions measured in Stripa groundwaters (from Andrews et al. 1982), which approximately match the profile expected if all the He produced is released into the groundwater, followed by diffusive loss from the groundwater itself ($D_{\text{water}}^{\text{He}} = 1 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ ($0.03 \text{ m}^2 \text{ a}^{-1}$)), suggesting that He loss from the minerals in which they were produced is not a significant rate-limiting step. Note that diffusion profiles for Ar through the solid rock would be even more extreme with virtually no change in Ar concentration until within a few cm of the surface.

gases are not produced in sufficient quantities to form a continuous fluid capable of advective flow alone. Crustal noble gases will remain in trace quantities in the local pore fluids. The transport of the noble gases to near surface systems is dependant therefore on the behavior of the fluid in the pore space. In a system that has no advective fluid flow, transport will occur by bulk diffusion, which in turn is controlled by the host rock character such as permeability, tortuosity, and the fluid occupying the pore space. When movement of fluid in the pore space occurs, these fluids will remove or carry away the crustal noble gases away from the site of release. These ‘carrier’ fluids may be produced within the rock formation by chemical change (e.g., hydrocarbon formation or metamorphic devolatilization), or be sourced externally to the system (e.g., an aquifer or magmatic fluid). In the former case, the crustal gases provide information about the source of the fluid, in the latter, information about the system through which they have passed (Ballentine et al. 2002, this volume).

Diffusion: A viable transport mechanism to degas the continental Crust?

Estimating the diffusivity of noble gases through crustal rocks—as opposed to

minerals—is non-trivial because of its strong dependence on both the rock type and temperature. At low temperatures volume diffusion through minerals or along grain boundaries is slow and the dominant diffusive route is expected to depend on the permeability and interconnectivity of the rock and the fluid medium (usually water). At high temperatures and in systems where permeability is low, noble gas diffusivity through whole rock will approach that of the volume diffusivity through the dominant mineralogy, and volume diffusion is expected to dominate the mechanism of gas transport. Several different approaches have been used to estimate the bulk diffusivity in a variety of systems (Table 9):

1. Direct measurement
2. Diffusion profiles in rocks
3. He fluxes in groundwaters

Two different direct measurement approaches have been used. Hussain (1996) incrementally heated bulk samples of the high U, high Th Carnmenellis granite. The increase in ^4He released with increasing temperature is used to calculate a diffusivity of $\sim 1 \times 10^{-15} \text{ cm}^2 \text{ s}^{-1}$ at 20 °C, greater than three orders magnitude faster than the constituent minerals of the granite. The high D_{He} is attributed by Hussain (1996) to the availability of recoil-implanted ^4He in cemented micro-cracks. Ohsumi and Horibe (1984) create a membrane of the material of interest (seafloor sediment cores) and evacuate one side of the material while applying a partial pressure of noble gas on the other. The amount of gas that diffused through the membrane in a given time period is then a function of the bulk diffusivity through the membrane. Note that diffusion of He out of the mineral in which it was produced is not involved in the latter diffusion measurements. The diffusivities of various sediments obtained by this method ranged from 13 to $30 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ for He and 3 to $6 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ for Ar measured at ambient temperature. These diffusion rates are only slightly lower than those obtained for He and Ar in water of 42×10^{-6} and $9 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ respectively (Jähne et al. 1987), suggesting that water-filled porosity may be the dominant diffusion medium in these seafloor sediments.

The pattern of ‘excess’ ^{40}Ar trapped in minerals can be used to estimate the bulk diffusivity through the rock. In theory, the concentration of ^{40}Ar in a mineral above its closure temperature should be close to zero. However, if the Ar is produced in the rock faster than it can diffuse out, then the mineral is likely to retain some ‘excess’ ^{40}Ar . The build-up of ^{40}Ar in a mineral above its closure temperature is a function of the relative rates of Ar production and diffusion out of the rock, not the mineral. Utilizing outcrop-scale gradients in ^{40}Ar concentrations in an amphibolite from the Simplon pass, Switzerland, Baxter et al. (2001) calculate a bulk Ar diffusivity of the order $10^{-11} \text{ cm}^2 \text{ s}^{-1}$ at a temperature of $\sim 500^\circ\text{C}$ (the conditions of their study). These results probably define the upper limit on the bulk Ar diffusivity in the upper crust (at least, for the immediate lithology of amphibolite) as diffusion rates are likely to decrease at lower temperatures. Studies by Foland (1979) and Scaillet (1996) also correlate ‘excess’ ^{40}Ar buildup with limited bulk diffusivity.

Torgersen (1989) formulates the steady-state He flux (F) as a function of the production rate (G) and diffusion geometry of the system. In an homogeneous slab

$$F/G = 1 - 8/\pi^2 f\{Dt/l^2\} \quad (31)$$

where D, l and t are the diffusivity, depth of crust and time respectively. The degassing of He into the Great Artesian Basin aquifer system, Australia, was shown to be close to whole crust production (Torgersen and Clarke 1985) and similar conclusions have been made for other sedimentary basins (e.g., Takahata and Sano 2000). From the solution of Dt/l^2 , ($t = 4.5 \times 10^9 \text{ yr}$; $l = 40 \text{ km}$) and conservatively estimating $F/G \geq 0.5$, Torgersen

(1989) estimate that the diffusivity of He through the crust would have to be $\geq 2 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. Torgersen (1989) scale this further assuming that 75% of the radioelements are in the upper 10 km of crust to obtain a $D_{\text{He}} \geq 1 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$. The effective bulk diffusivity of He through upper crustal rocks is unlikely to be this high, therefore transport of He by movement of the fluid through the crust is implied.

These last calculations provide an important perspective on the importance of diffusion as a transport mechanism over long length scales in the crust, as it is clear that the crust cannot be described as a static water filled porous medium over 4.5 Ga! Taking the diffusivity of He in ocean sediments at 20°C as an example, He can diffuse about 100 m per Ma. At the other extreme the limited evidence from the granite study suggests that diffusion rates through rock with limited porespace connectivity are many orders of magnitude lower and is not therefore a viable mechanism to efficiently transport noble gases from the deep crust to the surface.

Two very different mechanical regimes exist in the crust: (1) a shallow cool portion in which the rock is strong enough to allow a hydrostatic pressure gradient to be maintained but is likely episodically fractured following lithostatic over-pressuring (see below), creating a system of interconnected pores and pathways; and (2) a deeper hotter region where the rock has insufficient mechanical strength, except locally, to hold open the pores against the lithostatic pressure exerted by the overburden (e.g., Wood and Walther 1986). The transition from one domain to another depends on the rock composition and temperature gradient. In quartz-rich sediments and with a temperature gradient of 30°C/km, this transition occurs between 3- to 6-km depth (Rubey and Hubbert 1959; Fyfe et al. 1978) and is clearly seen in many sedimentary basins where fluid pressure becomes greater than hydrostatic at depths < 3 km (Mann and Mackenzie 1990). In crystalline rocks, this transition may occur over a broad temperature range, but typically occurs between 300-450°C (Manning and Ingebritsen 1999; Simpson 1999). For typical crustal gradients of 30°C/km, transition from the brittle to the ductile region occurs between 10- to 15-km depth.

Within the shallow crust, the diffusive length-scale of He relative to the rate of groundwater flow can be a significant factor in modeling noble gas loss from aquifer systems to the atmosphere. For example, loss by diffusion to the atmosphere can be modeled assuming a zero concentration boundary at the surface and a zero concentration initial condition (Andrews et al. 1989a):

$$C(z,t) = Gt_f [1 - \exp(-2z/\sqrt{(\pi Dt)})] \quad (32)$$

where $C(z,t)$ is the concentration at a given depth (z) and time (t), G is the production rate and t_f is the formation age. Some illustrative curves for diffusive loss of noble gases from the crust are shown in Figure 14, where the He concentrations in the water are perturbed on a ~100-m length scale as the boundary layer is approached. The length scale of the water affected by Ar diffusion will only be on the order of centimeters.

Differential release and transport of helium and argon

Low He/Ar ratios in fluids will result from release of noble gases (for example, by fracturing, chemical alteration or by raising the temperature) that have accumulated in minerals within an environment of preferential ^4He release. This general pattern has been observed in the KTB drill-hole, Germany, where rocks from the shallow portion of the drill-hole (<3000 m) have lower He/Ar ratios than those at the bottom of the drill-hole (3000 to 7000 m) (Bach et al. 1999; Drescher et al. 1998). Curiously, the variations in He/Ar in the KTB rocks result largely from high Ar concentrations near the top of the drill-hole; without isotopic information, it is not possible to determine if this is due to

variations in atmospheric contamination as opposed to preferential noble gas release from the crust. A more robust observation of this phenomenon are high $^4\text{He}/^{40}\text{Ar}^*$ ratios in shallow fluids from the Po basin, Italy. These high $^4\text{He}/^{40}\text{Ar}^*$, up to 120 compared to predicted ratios of ~ 5 , cannot result from extreme variations in $\text{U}+\text{Th}/\text{K}$ but must be due to preferential release of ^4He in the shallow crust (Elliot et al. 1993). Conversely, many fluids from the Great Artesian Basin (GAB), Australia (Torgersen et al. 1989), Permian Basin, Texas natural gases (Ballentine et al. 2001), The Trias aquifer within the Paris basin (Pinti and Marty 1995) and the Pannonian and Vienna Basins (Ballentine 1991; Ballentine et al. 1991) all have $^4\text{He}/^{40}\text{Ar}^*$ ratios close to the production value, consistent with their origin being from the deeper and hotter crust where diffusivities of Ar are sufficiently high to permit quantitative release of both He and Ar (Fig. 12).

In principle, the $^4\text{He}/^{40}\text{Ar}$ ratio in crustal fluids thus provides a tracer of the thermal conditions, or depth, from which the fluid was derived (Mamyrin and Tolstikhin 1984). In practice the regional release of ^4He and ^{40}Ar , and therefore the $(^4\text{He}/^{40}\text{Ar})$ ratio in the fluid will be dependent on the type, abundance and size distribution of the different minerals in which the respective noble gases are produced. For example, U in high heat producing rocks such as the Carnmenellis and Stripa granites is likely concentrated in fine grained ($<40\text{ }\mu\text{m}$) interstitial uraninite, facilitating ^4He release (Andrews et al. 1989a; Martel et al. 1990). One study from fluids trapped within the Swiss Alpine Palfris Marl, has put a limit on where the regional thermal transition from normal to high $^4\text{He}/^{40}\text{Ar}$ ratios occurs. Ballentine et al. (1994) use a stable isotopes and a constant $^{40}\text{Ar}^*$ concentration in CH_4 -rich fluid inclusions and CH_4 from associated borehole gases to argue that these fluids have a common origin. The trapped fluids are within carbonate

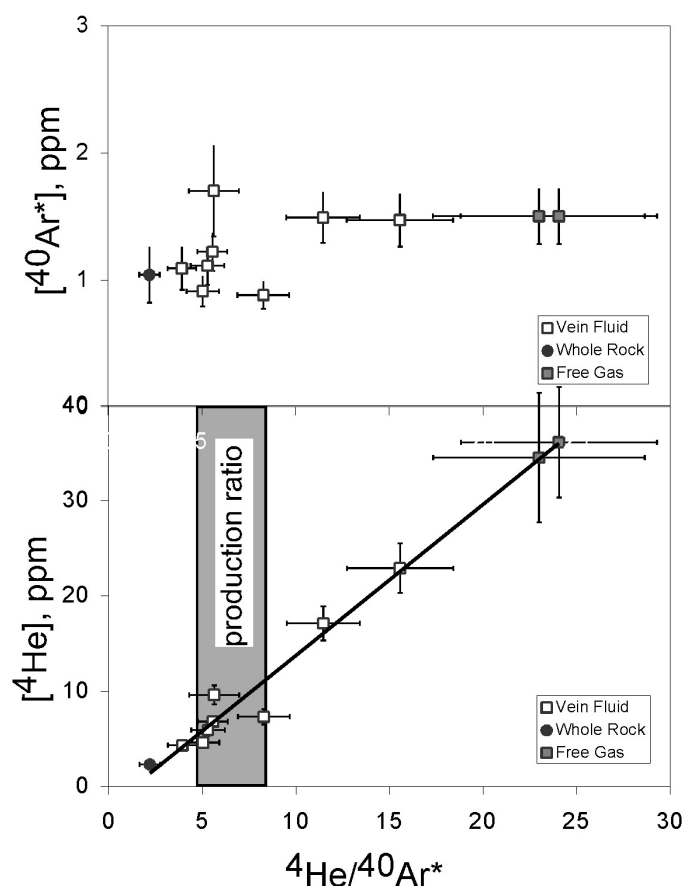


Figure 15. He and Ar concentrations are shown as a function of $^4\text{He}/^{40}\text{Ar}^*$ in the Palfris Marl (after Ballentine et al. 1994). Near-constant $^{40}\text{Ar}^*$ concentrations (all variation in $[^{40}\text{Ar}^*]$ is within 2σ of the mean: dashed line, top figure) are found in fluids trapped in vugs within the Palfris marl and in free borehole fluids from the same horizon. Stable isotopes and a constant $^{40}\text{Ar}^*$ show that these differently sited fluids are derived from the same source. In contrast, ^4He concentrations are $10\times$ higher in the free fluid than that trapped in the inclusions (lower figure) are likely due to addition of 'excess' ^4He to the free fluid. This is explained by ^4He , but not ^{40}Ar , diffusive loss into the fracture gas phase from the host marl. The transition to high $^4\text{He}/^{40}\text{Ar}$ ratios has occurred subsequent to the fluid formation and entrapment in the inclusions, at below 250°C and 6- to 8-km depth.

vein inclusions. Because of the low U+Th concentrations in the carbonate these are expected to have preserved the original fluid composition, with 6 out of 7 of the carbonates trapping fluids with $^4\text{He}/^{40}\text{Ar}^*$ ratios within error of the crustal production value (Fig. 15). In contrast the free gas phase shows a ^4He excess. This is explained by ^4He , but not ^{40}Ar , diffusive loss into the fracture gas phase from the host marl. The transition to high $^4\text{He}/^{40}\text{Ar}$ ratios has occurred subsequent to the fluid formation and entrapment in the inclusions. From stable isotopes, vitrinite maturity indicators and fluid inclusion formation pressures and temperatures, this occurred at below 250°C and 6- to 8-km depth.

Castro et al. (1998a,b) have modeled advective Ar and advective-diffusive He transport in a numerical simulation of different Paris Basin aquifer horizons and shown how these different transport mechanisms could account for the observed varying $^4\text{He}/^{40}\text{Ar}^*$ ratios. This interpretation is at variance with Pinti and Marty (1995) who have argued that impermeable layers act as efficient barriers for both He and Ar diffusion, and that the $^4\text{He}/^{40}\text{Ar}^*$ variation in the Paris basin results from advective mixing. Although diffusion may play a role in transporting He (but probably not the heavier noble gases) across boundary layers from one fluid system to another, in the absence of connected porosity (e.g., the crust below 3- to 15-km depth), diffusion as a significant mechanism of noble gas transport from the deep crust to shallow level systems can be neglected.

Accumulation and release: Fluid flow in the Crust

The steady-state model. There are many examples of crustal fluid systems that have concentrations of noble gases orders of magnitude higher than can be explained by local production, release and accumulation. One of the earliest examples is the series of studies on the noble gas concentration in groundwater from the Great Artesian Basin, Australia (Torgersen and Clarke 1985, 1987; Torgersen and Ivey 1985; Torgersen et al. 1989; Torgersen et al. 1992b; also see Bethke et al. 1999). An increase in concentration of noble gases with increasing age of the groundwater led to the development of the steady-state crustal degassing model. These workers argued that the flux of ^4He and $^{40}\text{Ar}^*$ into the base of the aquifer could be accounted for only if the entire crustal volume was releasing its radiogenic noble gases into the aquifer at the same rate as they were being produced. Similar rates have been inferred in other regional groundwater systems such as the Saijo Basin, Japan (Takahata and Sano 2000) and from natural gases reservoirs (Sano et al. 1986). Estimates of ^4He flux into lakes show similar high values (summarized by Kipfer et al. 2002, this volume), although varying over an order of magnitude in value. Hydrodynamic modeling by Castro et al. (1998a) implies a ^4He flux from the Paris basin equivalent to whole basin production, similar to the Great Artesian Basin. However, the model estimates are inconsistent with more direct estimates of He fluxes that use cyclic variations in the aquifer stable isotope record ($\delta^{18}\text{O}$ and δD) to estimate aquifer residence times (Marty et al 1993; Pinti and Marty 1995). The latter studies are used to indicate that the flux of ^4He from the basin center is likely to be lower by a factor of 10 than that produced within the aquifer.

Tectonic control of release. Both higher and lower fluxes are observed elsewhere: Within the Pannonian basin, a single gas field contains the volume of ^4He and $^{40}\text{Ar}^*$ produced in the entire crust beneath the sub-basin since the basin formation 15 Ma ago (Ballentine et al. 1991). This particular gas field shows evidence that the gases have been transported to the trapped site dissolved in an active groundwater system. A conceptual model in which basin extension releases deep stored noble gases, followed by groundwater focusing into the gas trap was proposed. Winckler et al (1997) measured exceptionally high ^4He and ^{40}Ar concentrations in brines from the Urania Basin of the

Mediterranean with ^4He concentrations in excess of $1 \times 10^{-3} \text{ cm}^3 \text{ STP } ^4\text{He g}^{-1} \text{ H}_2\text{O}$. These He concentrations could not be generated through any steady-state model.

A diffusive mechanism cannot account for significant noble gas mass transport from the deep crust. Nor, given the permeability of the crust, is it reasonable that there is continuous and widespread fluid loss from the deep crust. An average crust K age of 1 Ga (Hurley and Rand 1969) clearly indicates that although portions of the crust are extensively outgassed, regions of the crust must exist in which there are substantial accumulations of radiogenic and nucleogenic noble gases. Although at some point accumulations may reach a stage where discontinuous loss (both temporally and spatially) approximates the production rate, the rate of loss might be expected to be more strongly affected by large-scale tectonic events. For example, the regional strong thermal pulse caused by the formation of the Pannonian basin extensional system resulted in regional metamorphism, and the release of accumulated radiogenic noble gases at a rate far higher than they are produced in the crust (Ballentine et al. 1991), and furthermore shows clear evidence of magmatic involvement in the form of $^3\text{He}/^4\text{He}$ ratios well above crustal production values. Regions in which flux estimates are closer to theoretical steady-state values such as the Great Artesian Basin and Taiwan natural gases (Torgersen and Clarke 1987, 1987; Sano 1986) also show evidence of elevated $^3\text{He}/^4\text{He}$ ratios indicative of deep magmatic activity.

Similarly, orogenesis is also expected to result in regional metamorphism, fluid mobilization and regional degassing. Ar-Ar dating of these events clearly shows the resetting of mineral noble gas concentrations due to these processes (e.g., Vance et al. 1998), while a mass balance assessment of noble gases in the mineral and associated fluid phases in low grade Alpine metamorphism demonstrate extensive fluid loss from the system (Ballentine et al. 1994).

The concentration of radiogenic noble gases in fluids generated by regional fluid mobilization, and therefore their flux, will in turn be controlled by this history. In this respect, noble gas concentrations in the crust will also depend on the regional tectonic history. In summary:

- There is no a-priori reason, nor significant evidence, that regions of the crust do or should exhibit a uniform steady-state degassing flux.
- Crustal degassing is controlled by regional tectonic events resulting in metamorphism, such as orogeny and crustal extension.

The character of 'deep' crustal noble gases. $^4\text{He}/^{21}\text{Ne}/^{40}\text{Ar}$ ratios close to the predicted average crustal value are observed in many samples associated with either high radiogenic noble gas flux or magmatic fluids. These include natural gas samples from the Pannonian and associated Vienna basins (Ballentine 1991; Ballentine et al. 1991; Ballentine and O'Nions 1994), Red Sea brines (Winckler et al. 1997), magmatic CO_2 in the Val-Verde basin, Texas (Ballentine et al. 1991), Paris basin (Pinti and Marty 1995; Castro et al. 1998); and the Great Artesian Basin (Torgersen et al. 1989). Care must be taken within these studies to distinguish samples that have been subject to secondary fractionation process, such as phase fractionation. This can be established by identifying coherent fractionation in non-radiogenic gases such as atmosphere-derived noble gases introduced via the groundwater system (Ballentine et al. 1991). Similar studies that do not preserve predicted ratios can often be ascribed to shallow and fractionated radiogenic sources (e.g., Ballentine et al. 1994; Ballentine and Sherwood Lollar 2001). Nevertheless, the general inference is that samples ascribed to deep sources are accumulated, released and transported to near surface systems with minimal fractionation. This can occur only if:

1. Storage in the deep crust is efficient for all noble gases irrespective of site;
2. Release of different noble gases to the transporting media is not rate limited by diffusion;
3. Diffusion as a major mass transport mechanism is insignificant; and
4. Transport to the shallow systems is single phase.

HEAT AND HELIUM

The relationship between ^4He and heat

Radioactive decay of U-series elements accounts for >75% of the Earth's heat (O'Nions and Oxburgh 1983; Turcotte and Schubert 1982). U-series decay produces ^4He and Heat with a characteristic ratio of $3.7 \times 10^{-8} \text{ cm}^3 \text{ STP } ^4\text{He J}^{-1}$. However, processes in the mantle can efficiently fractionate He from heat, as the surface flux of ^4He from the mantle is only 5% of the equivalent Heat flux (see Graham 2002). The heat flow from the crust can be subdivided into that produced in the crust and that transported from the mantle through the crust (O'Nions and Oxburgh 1983). From estimates of the U + Th content of the continental crust, it would appear that only about half the continental heat flux was produced within the crust (O'Nions and Oxburgh 1983); this can be even lower in regions of crustal extension or tectonic deformation (Clark and Phillips 2000; Torgersen et al. 1992b). Heat and ^4He are produced in the mantle with the same ratio as in the crust. However, the convecting mantle has a near constant $^3\text{He}/^4\text{He}$ ratio ($\approx 1.2 \times 10^{-5}$), therefore the mantle also has a diagnostic $^3\text{He}/\text{heat}$ ratio that can be estimated from measurements of the He and Heat fluxes in oceanic basins. This can be used to tease apart fractionation of He and heat during transport across the mantle-crust boundary and fractionation of He and Heat that were produced in the crust.

Transport of mantle heat and helium through the crust

Given that heat transport through the crust is dominated by conduction, whereas He is transported into and through the crust primarily by advection (Bickle and McKenzie 1987), fractionation of He from heat is expected. Nevertheless, there does appear to be a global relation between the mantle heat flux and $^3\text{He}/^4\text{He}$ of the local crustal fluids (Polyak and Tolstikhin 1985): stable regions of crust have comparatively low conductive heat flow and radiogenic He (e.g., Russian Shield, Canadian Shield etc) whereas recently tectonized regions have high mantle heat flow associated with high $^3\text{He}/^4\text{He}$ (e.g., Pannonian Basin, Massif Central, Rhine, Yellowstone, Eastern Great Artesian Basin, East African Rift, Larderelo, Italy).

Polyak and Tolstikhin (1985) approached the problem of regional He isotope patterns by considering that He isotope compositions resulting from differences in radioelement or target nuclei concentrations will be averaged to some degree when entrained in crustal fluids. This enables a comparison between mantle heat flux through geotectonic units and He isotope composition. Averaging the $^3\text{He}/^4\text{He}$ ratios of fluids from particular geotectonic environments—primarily in Northern Europe and Asia, Polyak and Tolstikhin showed that there is a distinct temporal evolution to the $^3\text{He}/^4\text{He}$ ratio of groundwaters, with high $^3\text{He}/^4\text{He}$ ($>1 \times 10^{-6}$) limited to magmatic/tectonically 'young' regions typically <50 Ma (Fig. 16). Groundwaters from tectonically 'old' regions, such as shields, are characterized by low $^3\text{He}/^4\text{He}$ ratios indistinguishable from radiogenic production values (ca 10^{-8}). In a plot of $\log(^3\text{He}/^4\text{He})$ vs. $\log(\text{basement age})$ a smooth curve showing reducing $^3\text{He}/^4\text{He}$ with time is obtained (Fig. 17). Polyak and Tolstikhin observed that a plot of conductive heat flow vs. $\log(\text{tectonic age})$ has a similar form, implying there is a relation between conductive heat flow and $^3\text{He}/^4\text{He}$, i.e., mantle derived heat in the crust is accompanied by mantle derived He. This is shown in Figure

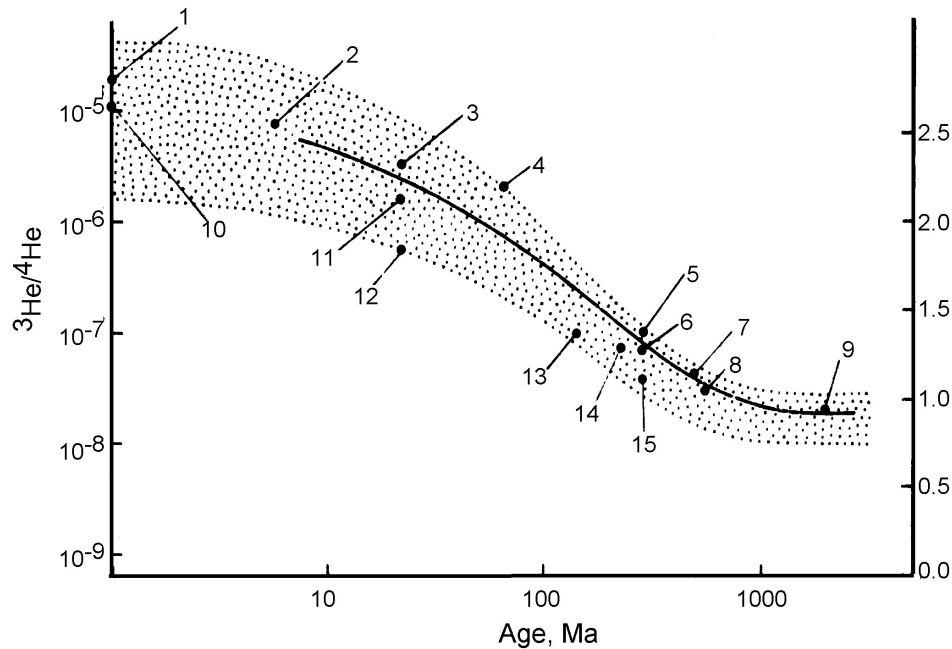


Figure 16. $^3\text{He}/^4\text{He}$ ratio as a function of age (Polyak and Tolstikhin 1985). The $^3\text{He}/^4\text{He}$ ratio of European groundwaters are plotted against the 'tectonic' age of the surrounding crust (stippled region; solid circles represent selected data). Adding more recent and/or global data increases the size of the envelope, particularly for young (<100 Ma) tectonostratigraphic units, but the relation between $^3\text{He}/^4\text{He}$ and age does not change significantly. The solid line is the conductive heat flux for the same tectono-stratigraphic units. The selected data are: 1: Iceland; 2: Eastern and Central Kamchatka; 3: Lesser Caucasus; 4: Western Kamchatka; 5: Scythian and Turanian Plates; 6: Mediterranean Foredeeps; 7, 15: Siberian Platform; 8: Baikalides; 9, 14: East European Platform; 10: Kuril Island Arc; 11: Greater Caucasus; 12: Trans-Caucasus Depression; 13: Vilyuy Syncline (From Polyak and Tolstikhin 1985, and references therein). The groundwater compositions better represent the compositions of the geotectonic units themselves as the groundwaters naturally average out the variability of the rocks. [Used by permission of Elsevier, from Polyak and Tolstikhin (1985), *Chemical Geology*, vol. 52, Fig. 8, p. 21.]

17 and takes the form:

$$^3\text{He}/^4\text{He} = (^3\text{He}/^4\text{He})_r (mQ - c) \quad (33)$$

where $(^3\text{He}/^4\text{He})_r$ is the radiogenic $^3\text{He}/^4\text{He}$ production ratio and m and c are constants; Q is the conductive heat flow. This relationship is only valid for mantle derived heat, with the lower intercept at the crustal $^3\text{He}/^4\text{He} = 10^{-8}$, where heat-flow is due to either crustal production or mantle heat unaccompanied by mantle He. The upper limit of the correlation assumes the mantle $^3\text{He}/^4\text{He} = 1.2 \times 10^{-5}$ and is due to the He/heat ratio of the mantle. The latter is unlikely to be exceeded during conductive heat flow because the effective diffusivity of heat through the crust is considerably greater than that of He. This value can, however, be exceeded if convective heat transfer occurs.

The relationship in some cases is less clear cut: for example, mantle He can be stored in the crust apparently without accompanying heat (Mirror Lake, Torgersen et al. 1994; Texas CO_2 well gases, Ballentine et al. 2001) and mantle heat can be transferred into the crust without affecting the $^3\text{He}/^4\text{He}$ composition of local groundwaters (North Atlantic, Oxburgh et al. 1986). Therefore, imperfect correlations are to be expected, but the considerable differences in He isotopic composition between the different potential He and heat sources means that useful information can be obtained from this approach.

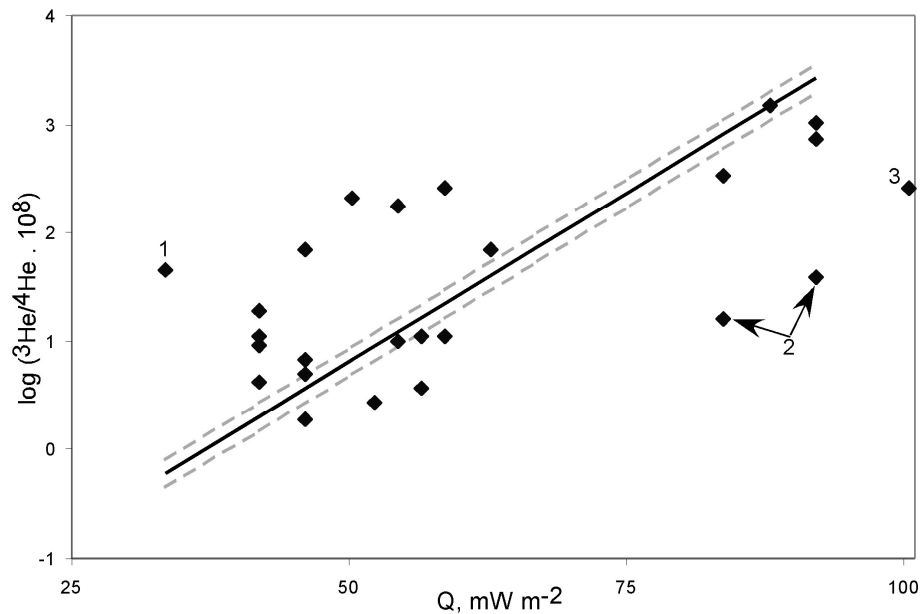


Figure 17. The relationship between $^3\text{He}/^4\text{He}$ and heat flux after Polyak and Tolstikhin (1985). The slope of the equation given by Polyak and Tolstikhin (1985) is shown by the solid line with error on the intercept indicated by the dashed line (slope = 6; intercept = -5.3 ± 0.2). The poor correlation between $\log(^3\text{He}/^4\text{He})$ and the heat flux shown here is consistent with mantle heat transported into the crust along with mantle He although with significant variability in the $^3\text{He}/\text{Heat}$ ratio supplied to the crust. It is noticeable that the outliers are limited to mobile tectonic belts where high geotherms likely result from vertical displacement: highlighted outliers are: 1. Eastern Black Sea; 2. Caucasus; 3. Baikal Rift. However, this correlation has been called into question by (Oxburgh and O’Nions 1987); further data show that high heat flow is also commonly found in regions that have low $^3\text{He}/^4\text{He}$ ratios. It would appear that while high $^3\text{He}/^4\text{He}$ ratios are usually accompanied by high heat flow, high heat flow is not always associated with high $^3\text{He}/^4\text{He}$.

Oxburgh and O’Nions (1987) observed that the helium isotopic compositions of some stable cratonic areas are low and uniform, but these can have variable—and quite high—heat fluxes (e.g., U.K. mainland and shelf has $^3\text{He}/^4\text{He} \leq 1 \times 10^{-7}$ but heat fluxes as high as 125 mW m^{-2}). Griesshaber and co-workers (Griesshaber et al. 1992, 1988) examined He-Heat relationships on a local scale and showed that there was no clear relationship between $^3\text{He}/^4\text{He}$ and heat flux in the extensional Rhine Graben. These observations contradict the findings of Polyak and Tolstikhin (1985), in part due to the differences in spatial scale of the two studies. A further complication is the variable amount of ^4He that may be stored regionally and mixed with the local fluids. A well-defined relationship between $^3\text{He}/^4\text{He}$ and heat over large geological provinces would seem surprising considering He and heat can readily be fractionated during relatively shallow processes.

Helium and Heat can be considered to have two components, one from the mantle and one produced in shallow crust:

$$Q_T = Q_M + Q_H \quad (34)$$

$$F_T = F_M + F_H \quad (35)$$

where Q and F are heat and helium fluxes respectively, subscripts T, M, and H are total, mantle, and crustal respectively fluxes respectively (strictly speaking, Q_H and F_H are the fluxes produced from a layer with a characteristic thickness ‘H’ and Q_M and F_M are the fluxes from below this layer). Given that F_M/Q_M should be constant, the relationship

between heat flow and $^3\text{He}/^4\text{He}$ will be more complex than suggested by Polyak and Tolstikhin (1985). If the crust dominates the fluxes of He and heat, $^3\text{He}/^4\text{He}$ will reduce with increasing heat flow. Conversely, in tectonic environments where crustal heat fluxes are unimportant, the $^3\text{He}/^4\text{He}$ ratio will increase with heat flux due to input of mantle He and Heat. The fact that there appears to be a positive correlation between $^3\text{He}/^4\text{He}$ and heat flux (Fig. 17) suggests that in regions where variations in $^3\text{He}/^4\text{He}$ can easily be measured (i.e., relatively high $^3\text{He}/^4\text{He}$) the mantle flux dominates. Subtle variations in $^3\text{He}/^4\text{He}$ in regions where crustal fluxes dominate are likely difficult to measure due to the low $^3\text{He}/^4\text{He}$ ratio of crustal dominated systems and so disappear in the noise.

Two different approaches have been used to investigate Helium and heat fractionation on the global scale by (Oxburgh and O'Nions 1987):

1. Assume the He flux from the mantle to the continents is the same as the He flux from mantle to oceans.
2. Assume that heat and He are supplied to the crust (from the mantle) at the same ratio that it is supplied to the oceans.

Both approaches are likely to lead to errors. On the one hand, it has been shown that there are continental regions where the mantle heat flux into the crust is lower than that of the oceans (Torgersen et al. 1992b, 1995). On the other, it is clear that the different transport properties of helium and heat through both mantle and crust will result in transfer of volatiles and heat to the crust with $^3\text{He}/\text{heat}$ ratios different from those of the mantle.

Fractionation of He from heat has also been observed in active and ancient hydrothermal systems from continental regions, confirming that the relationship between $^3\text{He}/^4\text{He}$ and mantle heat flow is little more than an association between heat and volcanism as opposed to a well constrained causal relationship that could be used (for example) to estimate heat fluxes from $^3\text{He}/^4\text{He}$ ratios. Extreme variations in $^3\text{He}/\text{Heat}$ (0.01 to $10 \times 10^{-12} \text{ cm}^3 \text{ STP } ^3\text{He J}^{-1}$) are observed in Icelandic hydrothermal fluids (Poreda and Arnorsson 1992) in contrast to submarine hydrothermal systems that have more constant $^3\text{He}/\text{Heat}$ ratios of between 0.1 and $1 \times 10^{-12} \text{ cm}^3 \text{ STP } ^3\text{He J}^{-1}$; (Jean-Baptiste et al. 1998). Modeling by Poreda and Arnasson (1992) show that fractionation by phase separation alone, which would result in a high- ^3He , low-heat vapor phase, cannot reproduce the observations. Instead, they suggest that combined fractional cooling and fractional degassing of the intrusive body is required to explain the data. From their models, He is preferentially lost from the magma such that (for each cooling-degassing increment): $(^3\text{He}/Q)_{\text{FLUID}}/(^3\text{He}/Q)_{\text{MAGMA}} \approx 20$, i.e., the magmatic body degasses twenty times faster than it cools. In the Icelandic case, convection is transferring mantle-derived volatiles into the crust more rapidly than mantle derived heat. This likely reflects the incompatible behavior of He during magmatic cooling rather than the intrinsic He and Heat transport properties.

Using the relationship derived by Turner and Stuart (1992) in order to estimate $^3\text{He}/Q_M$ (where Q_M is mantle derived heat) from measurements of noble gases in fluid inclusions, Burnard et al (1999) were able to estimate the $^3\text{He}/Q_M$ ratios of fluids related to Eocene gold deposition along the Ailaoshan Fault, Yunnan Province, China. Systematic variations along the fault showed that high $^3\text{He}/\text{heat}$ fluids were associated with the highest-grade gold ores and that the efficiency to extract fluids, and ^3He , from the parent intrusives decreased along with gold tonnages northward along the fault.

Transport of crustal heat and ^4He

The diffusion of heat through the crust ($10^{-2} \text{ cm}^2 \text{ s}^{-1}$) is considerably faster than that of He ($10^{-5} \text{ cm}^2 \text{ s}^{-1}$) (Torgersen et al. 1992a,b), therefore, crustal heat should be dissipated

more rapidly than crustal He. Torgersen et al (1992b) calculated typical time scales for heat and mass fluxes through a 10-km-thick crustal section to be ~0.7 and 80 to 800 Ma respectively. The disparity between heat and mass flux time scales will increase for thicker crustal sections.

Table 10. Helium and heat budgets from the GAB (Torgersen et al. 1992b) and the Pannonian Basin (Martel et al. 1989).

<i>Province</i>	<i>Total Heat flux mW m⁻²</i>	<i>Crustal Heat Flux mW m⁻²</i>	<i>⁴He flux cm³ STP m⁻²</i>	<i>Heat flux from ⁴He* mW m⁻²</i>	<i>Heat balance</i>
Central Shield	82.5	55.8	1.1 x 10 ⁻⁹	30	Additional 26 mW m ⁻² heat required
Eastern Australia	71.7	14.6	1.1 x 10 ⁻⁹	30	15 mW m ⁻² heat "lost"
Pannonian Basin	80	25 [♣]	3 x 10 ⁻⁹	81	51 mW m ⁻² heat "lost"

* U-series decay produces $3.7 \times 10^{-8} \text{ cm}^3 \text{ } ^4\text{He J}^{-1}$.

♣. error of $\pm 50\%$

Despite this prediction, the flux of ⁴He from the GAB, Australia, is roughly correlated with the flux of heat resulting from U-series decay within the basin (Torgersen et al. 1992b). The appropriate helium and heat budgets for the GAB and the Pannonian Basin are given in Table 10. Torgersen and Clarke (1985) demonstrated that the crustal He flux (F_H) from the GAB is $\sim 1.1 \times 10^9 \text{ cm}^3 \text{ STP } ^4\text{He m}^{-2} \text{ s}^{-1}$ ($3 \times 10^{10} \text{ atoms m}^{-2} \text{ s}^{-1}$). These results show that there is a discrepancy of about 50% in the He and heat fluxes, with a heat excess in the Central shield (relative to the heat flux accounted for by the ⁴He flux), while less heat (relative to the heat flux accounted for by the ⁴He flux) is coming out the Eastern shield. Minor ($\leq 25\%$) additional heat flux can result from K decay depending on the K/U ratio. However, considering the errors involved in these estimates, there is comparatively good agreement between He and heat fluxes for the GAB and Pannonian basins, which is not expected given their different transport properties. This may be explained if the loss of Heat and He is approaching steady state. Alternatively this may be explained if He transport out of the crust is faster than that predicted by diffusion. This is consistent with the observations of whole crust degassing, the inability of diffusion to achieve this over reasonable time scales, and the need for advective fluid transport of He out of the crust.

MAGMATIC NOBLE GASES IN THE CRUST

Tectonic control on magmatic fluid location

The general observation of Polyak and Tolstikhin (1985), that high ³He/⁴He is found in regions of high heat flow, restates the fact that mantle derived He in crustal fluids is found in regions of extension (Oxburgh et al. 1986; Oxburgh and O'Nions 1987). This is summarized in Table 11. In general, it can be seen that, with rare exceptions such as Mirror Lake, USA (Torgersen et al. 1995) and Val Verde Basin, Texas (Ballentine et al. 2001), fluids that have high ³He/⁴He ratios are from tectonically active regions and that the highest ³He/⁴He ratios are from regions undergoing active extension such as the Rhine Graben, Massif Central, East African Rift, or Great basin of the western United States. Active subduction also results in high ³He/⁴He groundwaters and geothermal fluids, e.g., Western Cordillera, Canada (Clark and Phillips 2000), Andes (Hoke et al. 1994) and the Western Pacific (Poreda and Craig 1989). Regions of crustal thickening

Table 11. Helium isotopes and concentrations in groundwaters, geothermal fluids, hydrocarbon and carbon dioxide wells.

<i>Basin</i>	<i>Tectonics setting</i>	<i>mean [⁴He] cm³ STP g⁻¹</i>	<i>³He/⁴He_{mean} x10⁻⁶</i>	<i>³He/⁴He_{min} x10⁻⁶</i>	<i>³He/⁴He_{max} x10⁻⁶</i>	<i>Analyses</i>	<i>References</i>
Europe							
Pannonian	extensional	7.56	0.76	0.15	1.97	27	Martel et al. (1989); Ballentine et al. (1991); Sherwood Lollar et al (1994)
Rhine	extensional	2.32	1.03	0.21	2.14	11	Hooker et al. (1985a)
Massif Central	extensional	0.30	2.36	0.42	3.92	6	Matthews (1987)
North Sea	extensional	8480 (oil)	0.46	0.28	0.54	18	Ballentine et al. (1996)
Lardarello, Italy	convergent	0.57	1.54	0.01	3.04	16	Hooker et al. (1985b)
Paris Basin	sedimentary	533.57	0.09	0.07	0.21	20	Marty et al. (1993)
Po	sedimentary	9.04	0.04	0.03	0.05	18	Elliott et al. (1993)
Mediterranean	sedimentary	641.56	0.08	0.08	0.09	8	Winckler et al. (1997)
Brine Basin							
Balkan Peninsula	convergent	4015.40	0.46	0.15	1.10	19	Piperov et al. (1994)
Alps	convergent	32.59	1.12	0.03	5.10	20	Marty et al. (1992)
Stripa, Sweden	shield	1744.00	0.01	0.01	0.01	3	Andrews et al. (1989a)
Iceland	extensional	78.26	21.71	7.70	40.32	47	Poreda et al. (1992)
Switzerland	convergent	1590.19	0.12	0.07	0.20	8	Tolstikhin et al. (1996)
Busko Spa, Poland	sedimentary/foredeep	179.60	0.04	0.05	0.03	9	Zuber et al. (1997)
Carpathians, Poland	extensional	26.00	0.53	0.49	0.57	2	Lesniak et al. (1997)
Vienna Basin	extensional	126.86	0.54	0.10	3.17	20	Ballentine and O’Nions (1992)
North America							
Sacramento	forearc	51.44	1.43	0.15	3.86	19	Poreda et al. (1986)
Great Basin, USA	extensional	1891.57	4.03	0.31	8.26	7	Welhan et al. (1988a,b)
West Texas	sedimentary	143.27	0.52	0.34	0.74	10	Ballentine et al. (2001); Zaikowski et al. (1987)
San Joaquin	forearc	61.05	1.31	0.15	3.86	24	Jenden et al. (1988)
Alberta Basin	sedimentary/forearc	152.42	0.20	0.01	0.69	20	Huyagon and Kennedy (1992)

Canadian Shield	shield											Bottomley et al. (1984)
Canadian Cordillera	sedimentary/forearc	15563.92	0.01	0.01	0.03	12						Clarke and Phillips (2000)
Rio Grande Rift	extensional	6.30	3.44	9.16	0.08	7						Smith and Kennedy (1985)
Long Valley	extensional	79.95	5.84	5.41	6.65	4						Hilton 1996; Welhan et al. (1988a,b)
		1582.73	5.66	0.63	7.70	11						
Australia, New Zealand												
GAB, Australia		49.53	0.27	0.04	1.13	34						Torgersen et al. (1985)
Taupo Volcanic Zone	convergent	18.89	7.35	4.93	8.48	10						Hulston and Lupton (1996)
Central America												
Macuspana, Mexico	extensional?	14.9	0.34	0.06	0.70	18						Battani (1999)
South America												
Central Andes	convergent	14.14	3.01	0.11	9.06	17						Hilton et al. (1993)
Africa												
East African rift	extensional	103.65	6.43	1.26	11.20	32						Darling et al. (1995)
Egypt	extensional	270.00	0.36	n/a	n/a	1						Sturchio et al. (1996)
Asia												
Tibet	convergent	1565.41	0.08	0.02	0.31	8						Hoke et al. (2000)
Songliao, China	extensional	380.39	4.69	0.95	6.99	6						Xu et al. (1995)
Wudalianchi China	extensional	304.60	3.80	1.67	4.72	6						Dai et al. (1996)
Indus Basin	Foreland	404.44	0.04	0.01	0.08	8						Battani et al. (2000)
Indian Craton	shield	23669.58	0.23	0.74	0.01	11						Minisalle et al. (2000)
Japan	Forearc	413.16	7.91	3.96	11.40	13						Nagao et al. (1981)

Note: Data compilations that do not include He and Ne concentrations have been excluded as it is not possible to assess atmospheric contamination. Direct samples of volcanic gases have been excluded

such as the Alps do not seem to result in high $^3\text{He}/^4\text{He}$ ratios (Marty et al. 1992): while there are regions of the Alps (and also of Tibet) where $^3\text{He}/^4\text{He}$ ratios are greater than likely production ratios, these do not appear to coincide with the main area of crustal thickening but rather with grabens or normal faulting at the periphery of collision (Marty et al. 1992; Hoke et al. 2000). Although transfer of other mantle-derived volatiles (such as Ne, Ar or CO_2) into crustal fluids also occurs in tectonically active regions, competing processes (such as radiogenic production in the crust) and high concentrations of these volatiles in the crust means their mantle-derived signatures are frequently obscured or complicated (e.g., Ballentine 1997): the isotopes of He remain the tool of choice for detecting mantle input into the crust.

The transfer of volatiles from the mantle into the crust is most efficient following melting of the mantle and intrusion of the melt into the crust: Watson and Brenan (1987) and Brenan and Watson (1988) have shown that fluid transport through the mantle is likely to be minimal in the absence of melt. This is also obvious in active volcanic regions where the mantle-derived ^3He flux decreases rapidly away from the volcanic centers (Sano and Wakita 1985). However, uncertainties in the crustal ^4He flux and the He concentration of the magmatic end-member make it difficult to estimate the mantle mass flux to the crust from He isotope ratios of crustal fluids. Torgersen (1993) showed that a flux as low as $10^3 \text{ atoms m}^{-2} \text{ s}^{-1}$ of ^3He would create a detectable change in the He isotope composition of the fluid, but depending on the efficiency of crustal ^4He outgassing, the range in $^3\text{He}/^4\text{He}$ observed in the fluid could span an order of magnitude (Fig. 18).

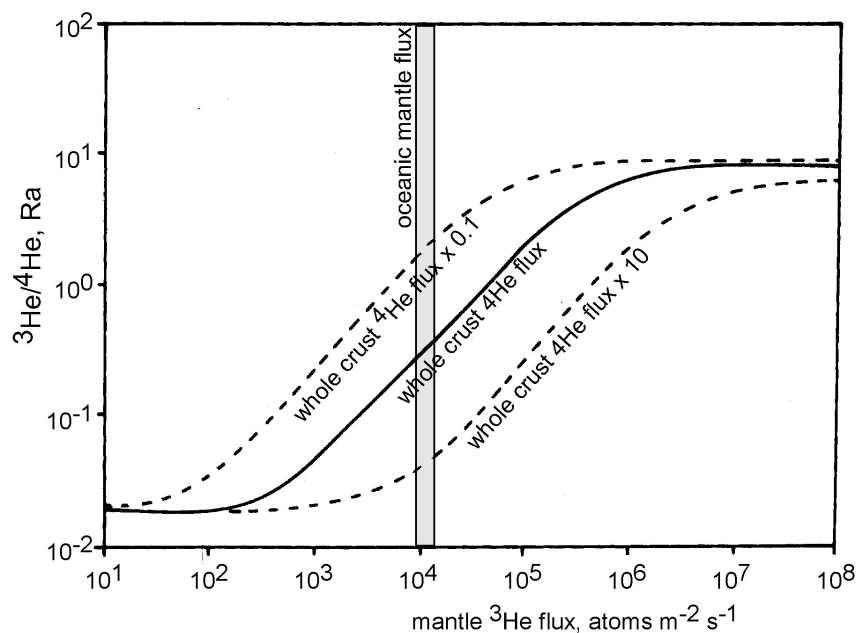


Figure 18. $^3\text{He}/^4\text{He}$ of groundwaters (after Torgersen 1993). The flux of ^3He from the mantle and ^4He from U series decay in the crust are the major contributions to the $^3\text{He}/^4\text{He}$ ratio of continental groundwaters. Because U concentration in the crust is relatively constant, the $^3\text{He}/^4\text{He}$ ratio of groundwaters is primarily a function of the ^3He flux from the subcontinental mantle. This figure illustrates the $^3\text{He}/^4\text{He}$ ratios expected in groundwaters for various mantle ^3He fluxes and crustal ^4He fluxes. The flux of ^3He from the mantle into the oceans is comparatively well constrained (gray bar): if the flux of ^3He from the sub-continental mantle was the same as that from the oceanic mantle, then all continental groundwaters would have $^3\text{He}/^4\text{He}$ ratios higher than the radiogenic production ratio, even if the flux of ^4He was 10 times greater than the production from U and Th decay. However, numerous groundwaters have He isotopic compositions that are indistinguishable from the radiogenic production ratio (Table 11), therefore the flux of ^3He under the continents is clearly lower than that under the oceans.

Fossil magmatic gases

It is clear that mantle-derived noble gases are stored in the crust as well as transported through it, as residual magmatic gases have been found trapped in fluid inclusions in ancient magmatic products [e.g., komatiites in Zimbabwe and Australia (Richard et al 1996); Greenland (Graham et al. 1998); Kola (Dauphas and Marty 1999; Marty et al. 1998), hydrothermal minerals (Burnard et al. 1999; Stuart et al. 1995; Stuart and Turner 1992), charnockites (Dunai and Touret 1993; Kamensky et al. 1990) and ultra-basic layered intrusions (Tolstikhin et al. 1992)]. While the concentrations of magmatic gases in these minerals can be surprisingly high (^3He concentrations can be similar to those trapped in basaltic glasses), these sites represent a small fraction of the crust and the trapped mantle noble gases are generally present in lower concentrations than radiogenic noble gases produced in the crust.

Magmatic fluids can also be preserved. For example, fossil mantle gases are preserved in the gas fields of western Texas (Ballentine et al. 2001). $^3\text{He}/\text{CO}_2$ ratio in natural gas wells in the Val Verde basin increase toward the Marathon Thrust Belt that defines the southern edge of the basin. These ratios are within the range predicted for magmatic gases, and the variation has been ascribed to Rayleigh fractionation during magma degassing. Ballentine et al (2001) combine this with a simple filling model to define the direction of the magma source. Combined with the observed spatial relationship between CH_4 and CO_2 this suggest that the magmatic gases pre-dated hydrocarbon generation in the basin. Hydrocarbon generation is known to have initiated at around 280 Ma, requiring that the mantle derived CO_2 and helium has been stored in crustal traps for a period of ~ 300 Ma. Similarly, Marty et al. (1993) detected a ^3He mantle component in deep waters in the Paris basin and attributed this to Hercynian magmatism intruding the basement at >250 Ma. Young groundwaters of the eastern United States also have $^3\text{He}/^4\text{He}$ ratios (up to 1.2 Ra) that require a fossil magmatic helium source (Torgersen et al. 1994; Torgersen et al. 1995). No volcanism has occurred in the eastern United States for >95 Ma requiring crustal storage of the He for at least this length of time. Modeling shows that, given the U concentration of the crust, high gas concentrations are required in order to preserve the high $^3\text{He}/^4\text{He}$ ratios. Although extant fluids from the mantle or those released from magmatic rocks can perturb local noble gas isotopic signatures, this source of magmatic fluids does not represent a significant mass within the crustal system.

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